



CME 3204

Data Communications and Computer Networks

Midterm Project – 2025-2026 Spring

Metropolitan Area Network Simulation

Project Report

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Abstract

This report presents the design and simulation of a Metropolitan Area Network (MAN) developed as part of the CME 3204 Data Communications and Computer Networks project. The network comprises two geographically distinct branch offices interconnected through an Internet Service Provider (ISP) infrastructure using Cisco Packet Tracer simulation software. The first branch hosts three facilities, including a server farm equipped with Web, FTP, Mail, DHCP, and DNS servers. The second branch also contains three facilities supporting a variety of end-user devices, including workstations, laptops, smartphones, and tablets. Seven mandatory simulation scenarios and two additional custom scenarios have been implemented and analyzed, covering HTTP browsing, email communication, FTP file transfer, VoIP conferencing, ICMP ping, and SSH remote access. The report details the network architecture, IP addressing scheme, simulation methodology, and traffic analysis results.

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1. Introduction

Computer networking has become a fundamental infrastructure layer for modern organizations. Designing, simulating, and validating a network before physical deployment significantly reduces costs and errors. This project focuses on building a Metropolitan Area Network (MAN) simulation using Cisco Packet Tracer, a discrete-event network simulator widely used in academic and professional settings. The MAN connects two branch offices within the same metropolitan area through an ISP, supporting a variety of services including web browsing, email, FTP, VoIP, and SSH.

1.1 Project Definition and Problem Formulation

The core objective of this project is to design, configure, and simulate a MAN consisting of two branch offices, each with multiple facilities housing diverse user types and server infrastructure. The network must reliably support real-world services such as HTTP, FTP, SMTP, VoIP, DNS, DHCP, and SSH. The design challenge involves balancing hardware resources, IP address allocation, routing protocols, and service configurations while adhering to the given specifications.

Specific challenges include ensuring seamless cross-branch communication, proper DHCP address assignment, correct DNS resolution, stable VoIP media transport, and secure SSH access, all within a simulated environment that faithfully reflects real-world network behavior.

1.2 Purpose and Motivation of the Project

The primary motivation behind this project is to apply theoretical networking concepts to a practical, hands-on simulation exercise. Organizations increasingly depend on reliable MAN infrastructure for inter-branch operations, cloud services, and real-time communication. Understanding how to architect such networks, configure routing and switching, and troubleshoot service failures is a critical competency for network engineers.

From a business perspective, a well-designed MAN reduces operational costs by enabling centralized server management, improves productivity through seamless cross-site communication, and enhances resilience through redundant routing paths. The risks include misconfiguration leading to service outages, IP address conflicts, and routing loops, all of which this simulation exercise aims to identify and resolve before a hypothetical real-world deployment.

1.3 Related Work

Network simulation and modeling have been extensively studied in the academic literature. Issariyakul and Hossain [2] provide a comprehensive foundation in network simulation methodologies using NS2, introducing discrete-event simulation concepts applicable to any simulation platform including Cisco Packet Tracer. Maria [1] introduced general principles of modeling and simulation relevant to computer network design, emphasizing the importance of simulation as a risk-free validation tool. Several academic

studies have used Cisco Packet Tracer specifically for MAN and WAN design validation, demonstrating its effectiveness in teaching and research contexts.

2. Method and Simulation

2.1 Simulation and Modeling Concepts

A Metropolitan Area Network simulation is an example of a discrete-event system (DES). In a DES, the system state changes only at specific points in time, called events, such as the arrival of a packet, the completion of a transmission, or the expiration of a timer. Between events, the system state remains unchanged, making DES efficient for modeling communication networks where activities are driven by packet arrivals and departures.

Modeling a network in simulation, rather than implementing it physically, offers several advantages: it eliminates hardware procurement costs, enables rapid what-if analysis, allows testing under fault conditions without risk, and provides reproducible experiments. Challenges of simulation include model validity (whether the simulation accurately represents reality), abstraction limitations, and the absence of real-world noise and hardware variation.

2.2 Simulation Environment and Tool

Cisco Packet Tracer is the simulation environment used in this project. It is a free, cross-platform network simulation tool developed by Cisco Systems, widely used in Cisco Networking Academy courses. Packet Tracer supports a range of Cisco devices including routers, switches, wireless access points, and end-user devices.

Its modeling approach is based on discrete-event simulation, where packets traverse a virtual network according to configured protocols and topologies. The tool supports both real-time mode and simulation mode, where the latter allows step-by-step packet tracing with PDU inspection at each OSI layer.

Key capabilities include support for IPv4/IPv6 addressing, static and dynamic routing (RIP, OSPF, EIGRP), VLANs, NAT, DHCP, DNS, HTTP, FTP, SMTP/POP3, VoIP (using SCCP), SSH, and wireless (WPA2) configurations. Limitations include the inability to simulate all vendor-specific features and some advanced protocols not included in its built-in library.

In Cisco Packet Tracer, the network is not "programmed" using traditional high-level software languages. Instead, it is built and configured using a combination of Graphical User Interface (GUI) actions and Command Line Interface (CLI) commands.

- **Programming (Configuration):** Physical topology design, device placement, cabling, and end-device settings are established through the GUI. However, the core network logic, such as static/dynamic routing, VLAN assignments, SSH security parameters, and VoIP (Call Manager Express) setup, is programmed

directly into the routers using Cisco Internetwork Operating System (IOS) commands via the CLI. Server functionalities (DNS, DHCP, Web, FTP, Mail) are programmed using their respective GUI service panels.

- **Running the Simulation:** The environment provides two execution modes to run the network. Realtime Mode runs continuously, mimicking actual physical hardware operations, and is used for initial configuration and basic connectivity tests. Simulation Mode operates as a discrete-event execution environment and is utilized for detailed traffic analysis. In Simulation Mode, the time is paused and controlled by the user, allowing packet-by-packet (PDU) tracing across the network topology, event logging, and deep inspection of OSI layer data during specific test scenarios.

The primary components and modules used in this project are Cisco 2911 router , Cisco 2960 switch, generic server objects (configured for Web, FTP, Mail, DHCP, and DNS services), Cisco Wireless Access Points (WAPs), and end-user devices (PCs, laptops, smartphones, tablets).

2.3 Network Design Requirements

The MAN topology must satisfy the following high-level architectural requirements derived from the project specification:

- Two geographically separated branch offices connected via an ISP.
- Each branch must have a minimum of two routers for redundancy or routing.
- Branch 1 contains three facilities with mixed user types and a centralized server farm.
- Branch 2 contains three facilities with workstations, wireless devices, and mobile users.
- All users must access appropriate services (Web, FTP, Email, VoIP, SSH) based on their facility specifications.

The table below summarizes the network components:

Component	Branch 1	Branch 2	Total
Branch Routers + ISP	2	2	5
Switches	3	3	6
PCs / Workstations	3 + 6 + 0 = 9	5 + 5 + 5 = 15	24
Laptops (Wireless)	3	5	8

Component	Branch 1	Branch 2	Total
Smartphones/Tablets	3	5 + 2 + 2 = 9	12
Web Servers	10	—	10
FTP Servers	4	—	4
Mail Server	1	—	1
DHCP Server	1	—	1
DNS Server	1	—	1

Table 1: Network Component Summary

2.4 Requirement Analysis

2.4.1 Functional Requirements

The following functional requirements are defined for each service:

- HTTP/HTTPS: All user types in facilities specified by the project must be able to browse web pages hosted on Web servers located in Branch 1, Facility 3.
- FTP: Workstation users in Branch 1 (Facility 2) and Branch 2 (Facility 2) must be able to upload and download files to/from FTP servers.
- SMTP/POP3 (Email): Users in Facilities 1 and 2 of both branches must be able to send and receive emails through the central Mail server in Branch 1, Facility 3.
- VoIP: Two workstations in Branch 1, Facility 2 must support VoIP conferencing, requiring a Cisco Call Manager Express (CME) or equivalent configuration.
- DNS: All devices must resolve domain names through the centralized DNS server.
- DHCP: Devices must receive IP addresses dynamically from the DHCP server in Branch 1, Facility 3.
- SSH: Mobile device users in Branch 2, Facility 3 must be able to remotely access Web servers via SSH.

2.4.2 Performance Requirements

The network must support simultaneous connections from all users described in the specification (a minimum of 24 workstations, 13 wireless devices, and 10 mobile devices). Routing must ensure inter-branch connectivity with acceptable latency. VoIP traffic requires low latency and jitter, which must be considered in the routing design.

To prevent internal bottlenecks and support high-bandwidth applications like FTP and HTTP, the intra-facility Local Area Network (LAN) connections utilize FastEthernet (100 Mbps) and GigabitEthernet (1 Gbps) interfaces using copper straight-through cabling.

2.4.3 Constraints

- All simulation must be performed within Cisco Packet Tracer.
- Each branch must use at least two routers.
- IP addressing must follow a systematic, documented scheme.
- The final design file must be error-free and fully functional.

2.5 Definitions of the System/Model

The following assumptions underpin the simulation model:

- All end-user devices obtain IP addresses via DHCP unless statically configured.
- The ISP infrastructure is represented by a single dedicated router, which acts as the backbone simulating MAN (Metropolitan Area Network) connectivity between the two branch offices.
- Routing between branches is achieved through dynamic routing protocols.
- Wireless connectivity is achieved through access points using infrastructure mode (BSS).

The network utilizes a hierarchical Core-Edge architecture. Topologically, the internal LANs within each facility are built on physical star and extended-star designs, while the external WAN consists of point-to-point serial links connecting each branch's Edge router to a single central ISP backbone router. Branch 1 uses its Core and Edge routers to manage three facilities, including a dedicated VoIP segment (Facility 2) and a centralized Server Farm (Facility 3) that provides core services. Branch 2 mirrors this routing structure, hosting three facilities dedicated entirely to end-user devices (workstations, wireless laptops, and mobile devices) that access the centralized resources located in Branch 1.

The network simulation handles multiple data types dictated by the active application protocols, including HTML text for web browsing, binary files for FTP transfers, encrypted data streams for SSH sessions, and UDP datagrams for DNS, DHCP, and VoIP traffic. The primary data sources are the centralized servers located in Branch 1, Facility 3, which respond to client requests and provide core services to the entire network. The system's hardware is categorized into two main device types: managed devices (routers, switches, and access points) that form the infrastructure responsible for traffic forwarding, and host devices (wired workstations, wireless laptops, smartphones, and IP phones) that act as the end-user endpoints initiating data transmission.

Dynamic Host Configuration Protocol (DHCP) Operations:

In this network model, IP address management is centralized to ensure scalability and administrative efficiency. A dedicated DHCP server is deployed within the Server Farm located in Branch 1, Facility 3. Because routers natively drop Layer 2 broadcast

packets, end-user devices situated in different physical locations (such as the wired workstations and wireless clients in Branch 2) cannot reach the DHCP server directly. To resolve this, the Core and Edge routers are configured with DHCP Relay Agents (using the `ip helper-address` command). When a newly connected client broadcasts a DHCP Discover packet, the local router intercepts it, encapsulates it into a unicast UDP packet, and routes it across the OSPF-configured ISP backbone directly to the centralized DHCP server. The server then allocates an appropriate IPv4 address, subnet mask, and default gateway from the specific pool, ensuring seamless and automated network integration for all end devices.

Domain Name System (DNS) Operations:

To facilitate user-friendly access to internal resources, a centralized DNS server is concurrently hosted within the Branch 1, Facility 3 Server Farm. The fundamental role of this server in the simulation is to map human-readable domain names to the static IP addresses of the internal application servers (Web, FTP, and Mail servers). When a user in any facility initiates an application request using a URL rather than an IP address, the host device generates a DNS Query encapsulated in a UDP datagram targeting port 53. This query is routed across the MAN infrastructure to the DNS server, which consults its internal records and returns the corresponding IP address. This resolution process is critical for the functioning of application-layer services, as it allows dynamic routing protocols to establish end-to-end TCP connections without requiring users to memorize internal IP schemes.

2.5.1 Routing Protocols

In this MAN simulation, Dynamic Routing using OSPF (Open Shortest Path First) was implemented via the Command Line Interface (CLI) across the branch routers and the ISP backbone. OSPF, an industry-standard link-state routing protocol, was selected due to its scalability, fast convergence, and efficiency. Although the current topology consists of two branches connected via an ISP, a Metropolitan Area Network must be designed with future expansion and fault tolerance in mind.

2.5.2 Wireless Security Standard

Wireless access across the network is secured using the WPA2-PSK (Wi-Fi Protected Access II - Pre-Shared Key) authentication standard. This ensures that all wireless communications from laptops, smartphones, and tablets are heavily encrypted using AES (Advanced Encryption Standard), protecting the corporate network from unauthorized access and eavesdropping.

To logically separate and easily identify the wireless access networks across different physical locations, the following Service Set Identifiers (SSIDs) were configured on the respective Access Points:

- **Branch 1 - Facility 1:** B1_F1_Wifi

- **Branch 2 - Facility 1:** B2_F1_Wifi
- **Branch 2 - Facility 2:** B2_F2_Wifi
- **Branch 2 - Facility 3:** B2_F3_Wifi

2.5.3 IP Address Space Table

The IP addressing scheme follows a systematic subnetting approach to minimize waste and simplify management. The table below defines the planned address space:

Network Segment	Network Address	Subnet Mask	Gateway
Branch 1 – Facility 1	192.168.10.0	255.255.255.0	192.168.10.1
Branch 1 – Facility 2	192.168.20.0	255.255.255.0	192.168.20.1
Branch 1 – Facility 3 (Server Farm)	192.168.30.0	255.255.255.0	192.168.30.1
Branch 2 – Facility 1	192.168.40.0	255.255.255.0	192.168.40.1
Branch 2 – Facility 2	192.168.50.0	255.255.255.0	192.168.50.1
Branch 2 – Facility 3	192.168.60.0	255.255.255.0	192.168.60.1
ISP Link – B1_Edge_Router	10.1.2.0	255.255.255.0	—
ISP Link – B2_Edge_Router	10.2.2.0	255.255.255.0	—
B1_Core_Router – B1_Edge_Router	10.1.1.0	255.255.255.0	—
B2_Core_Router – B2_Edge_Router	10.2.1.0	255.255.255.0	—

Table 2: IP Addressing Scheme

2.6 Simulation Elements

The simulation elements are categorized according to the discrete-event simulation framework described by Maria [1] and Issariyakul and Hossain [2]:

- **System Entities:** The primary entities consist of the hardware and logical components of the network. This includes end-user devices (workstations, laptops, smartphones, IP phones), infrastructure devices (routers, switches, wireless access points), centralized servers (Web, FTP, Mail, DHCP, DNS), and the Protocol Data Units (PDUs) such as frames and packets flowing between them.
- **System State Variables:** These variables define the condition of the network at any specific moment in the simulation. Key state variables include the operational status of network interfaces (up/down), channel status (idle or busy), OSPF routing table convergence states, and the availability of the DHCP IP address pools.
- **Input Variables:** The input variables are driven by the application-layer traffic generated during the test scenarios. These include packet generation rates triggered by user actions, data payload sizes (such as FTP file sizes or HTTP page requests), and continuous transmission rates defined by VoIP codecs.
- **Resources:** The system's capacity is constrained by its physical and logical resources, primarily consisting of the network link bandwidths (100 Mbps FastEthernet, 1 Gbps GigabitEthernet, and specific serial clock rates), device port availability, and the processing/buffer queue capacities of the routers and switches.
- **Activities and Events:** The discrete events that drive the simulation forward include packet generation, propagation delay over links, queuing at router interfaces, MAC address resolution (ARP broadcasts), TCP three-way handshakes, OSPF link-state updates, and the ultimate arrival or dropping of packets.

3. Traffic Analysis and Simulation Results

This section presents a comprehensive evaluation of the network's performance and functional integrity through various discrete-event simulation scenarios. The primary objective of these tests is to verify that the configured Metropolitan Area Network (MAN) correctly handles multi-protocol traffic, ensures end-to-end reachability, and maintains data encapsulation standards across different network segments.

3.1 Scenario 1 – Wireless User Reads Email and Browses the Web (B2-F1)

Network Functionality

A wireless laptop user in Branch 2, Facility 1 initiates an HTTP request to a Web server in Branch 1, Facility 3, and also checks incoming email via POP3 from the Mail server. The DHCP-assigned IP of the laptop is used as the source. The packet traverses the

wireless access point, the local switch, the branch router, the ISP link, and arrives at the destination server.

1) Application & PDU Parameters:

To simulate this scenario, two applications were used on the source laptop. First, the "Web Browser" application was opened, and the Web Server's IP address (or URL) was entered to generate an HTTP GET request (Port 80). Secondly, the "Email" application was opened and the 'Receive' button was clicked to initiate a POP3 session (Port 110) towards the Mail Server.

The generated PDUs encapsulate data through the OSI layers before transmission. Below are the key OSI layer activities observed in the PDU Information window at the source device:

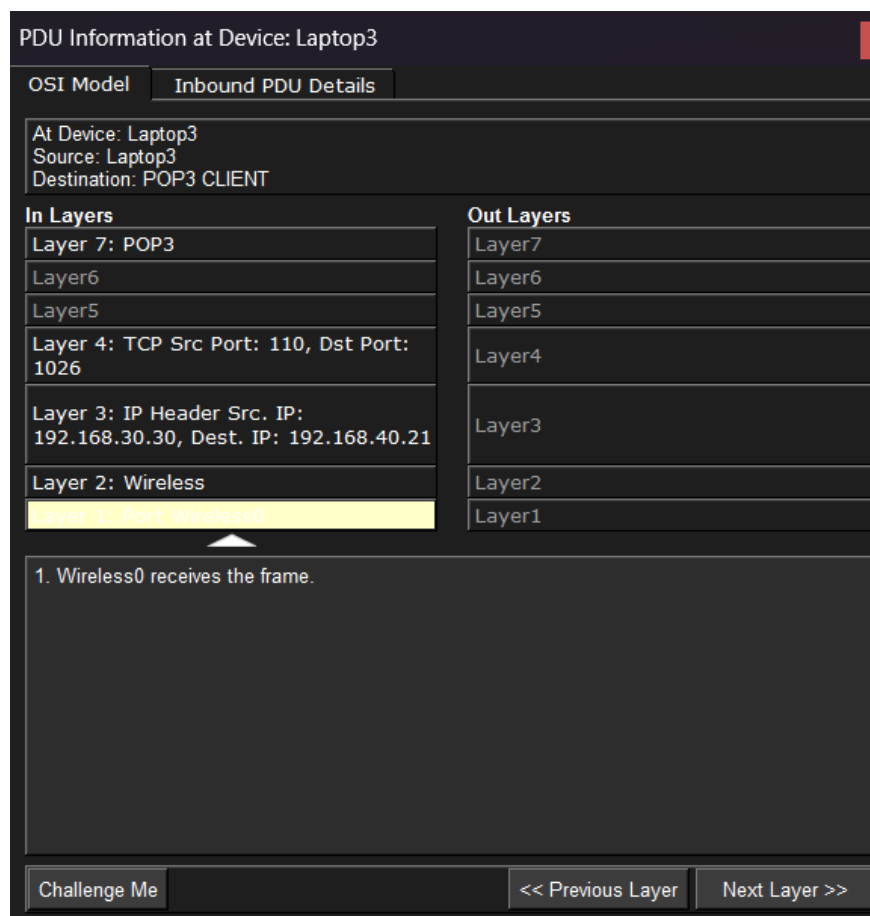


Figure 3.1.1: OSI Model at Device: Laptop3(POP3)

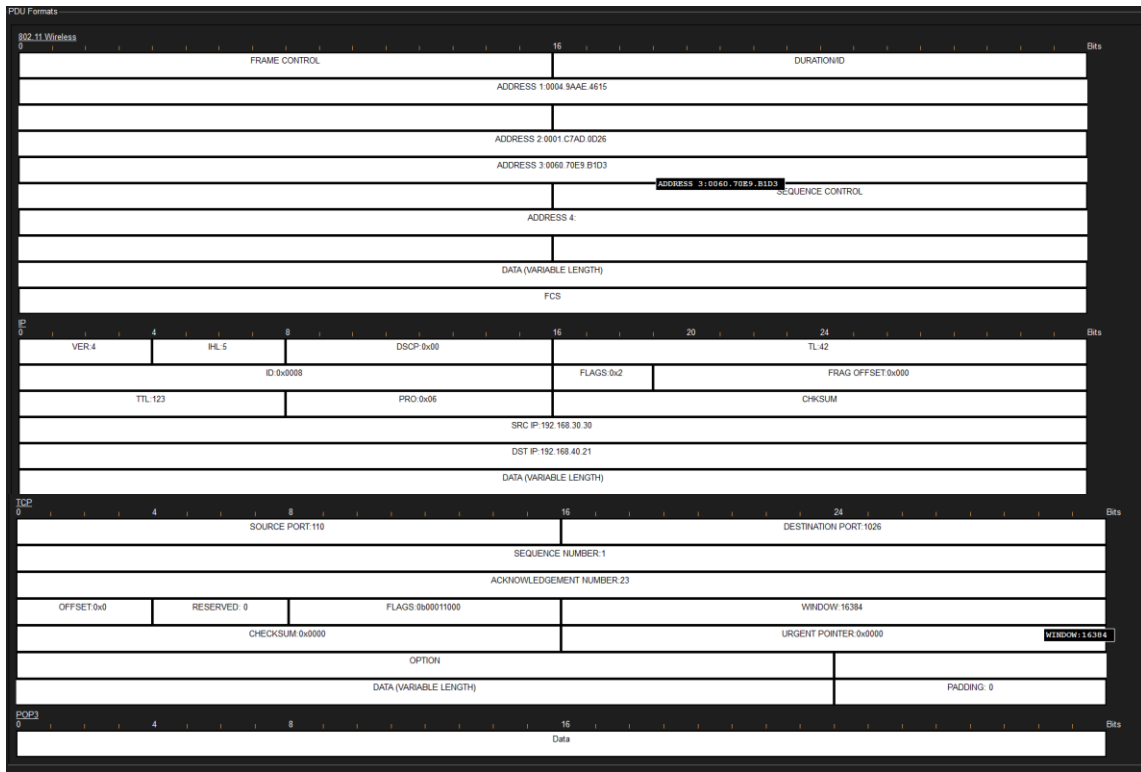


Figure 3.1.2: PDU Information at Device: Laptop3(POP3)

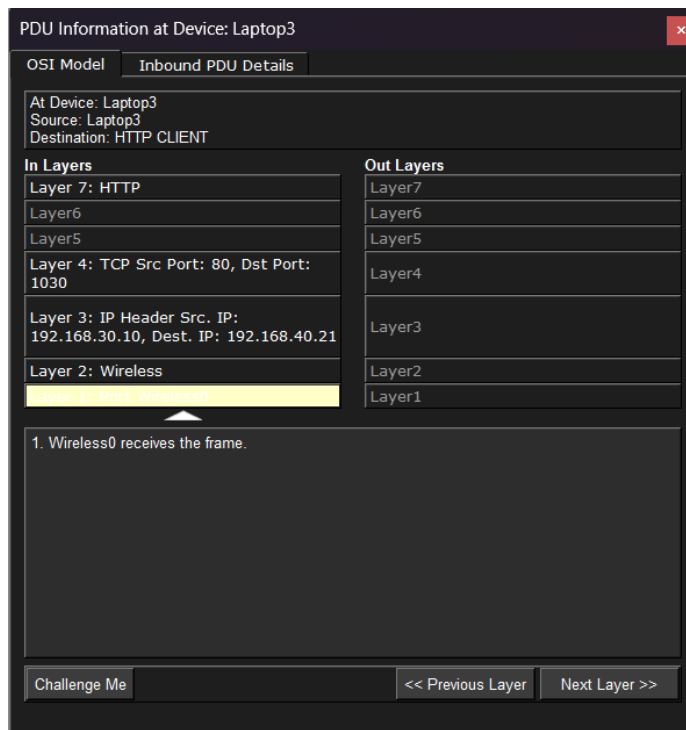


Figure 3.1.3: OSI Model at Device: Laptop3(HTTP)

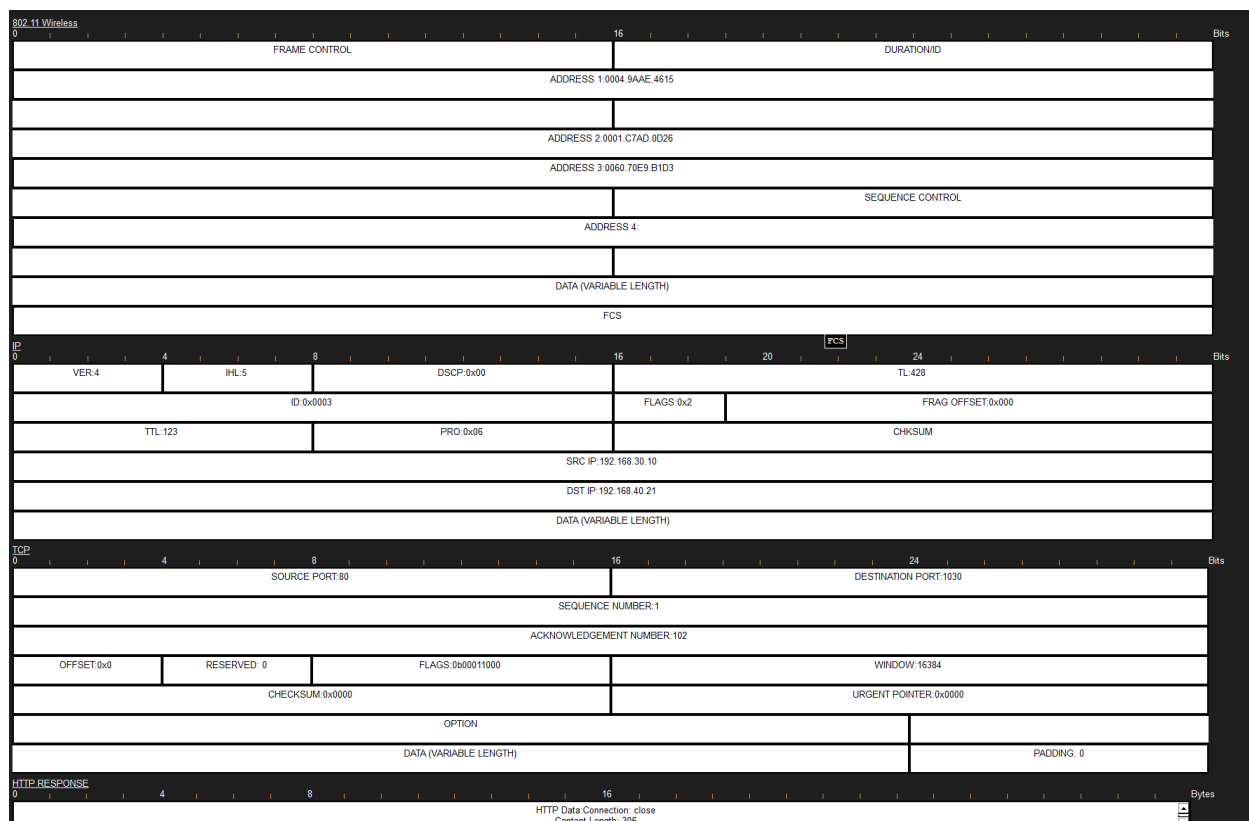


Figure 3.1.4: PDU Information at Device: Laptop3(HTTP)

2) Relevant Events Lists:

The following table traces the HTTP packet's hop-by-hop journey from the web server to destination Laptop.

Event List				
Vis	Time(sec)	Last Device	At Device	Type
	0.352	--	Laptop3	HTTP
	0.358	--	Laptop3	HTTP
	0.359	Laptop3	Access Point1	HTTP
	0.360	Access Point1	B2_F1_Switch	HTTP
	0.361	B2_F1_Switch	B2_CoreRouter	HTTP
	0.362	B2_CoreRouter	B2_EdgeRouter	HTTP
	0.363	B2_EdgeRouter	ISP_Router	HTTP
	0.364	ISP_Router	B1_EdgeRouter	HTTP
	0.364	--	Access Point1	HTTP
	0.365	Access Point1	Tablet PC4	HTTP
	0.365	Access Point1	Laptop4	HTTP
	0.365	Access Point1	Laptop3	HTTP
	0.365	Access Point1	Laptop6	HTTP
	0.365	Access Point1	Laptop7	HTTP
	0.365	Access Point1	Tablet PC0	HTTP
	0.365	Access Point1	Tablet PC2	HTTP
	0.365	Access Point1	Tablet PC1	HTTP
	0.365	Access Point1	Tablet PC3	HTTP
	0.365	Access Point1	Laptop5	HTTP
	0.365	B1_EdgeRouter	B1_CoreRouter	HTTP
	0.366	B1_CoreRouter	B1_F3_Switch	HTTP
	0.367	B1_F3_Switch	WEB1	HTTP
	0.368	WEB1	B1_F3_Switch	HTTP
	0.369	B1_F3_Switch	B1_CoreRouter	HTTP
	0.370	B1_CoreRouter	B1_EdgeRouter	HTTP
	0.371	B1_EdgeRouter	ISP_Router	HTTP
	0.372	ISP_Router	B2_EdgeRouter	HTTP
	0.373	B2_EdgeRouter	B2_CoreRouter	HTTP
	0.374	B2_CoreRouter	B2_F1_Switch	HTTP
	0.375	B2_F1_Switch	Access Point1	HTTP
	0.376	Access Point1	Tablet PC4	HTTP
	0.376	Access Point1	Laptop3	HTTP
	0.376	Access Point1	Laptop4	HTTP
	0.376	Access Point1	Laptop5	HTTP
	0.376	Access Point1	Laptop6	HTTP
	0.376	Access Point1	Laptop7	HTTP

Figure 3.1.5: Event List HTTP

The following table traces the POP3 packet's hop-by-hop journey from the Mail Server to the destination Laptop.

Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.050	--	Access Point1	POP3
	0.051	Access Point1	Tablet PC4	POP3
	0.051	Access Point1	Laptop4	POP3
	0.051	Access Point1	Laptop3	POP3
	0.051	Access Point1	Laptop6	POP3
	0.051	Access Point1	Laptop7	POP3
	0.051	Access Point1	Tablet PC0	POP3
	0.051	Access Point1	Tablet PC2	POP3
	0.051	Access Point1	Tablet PC1	POP3
	0.051	Access Point1	Tablet PC3	POP3
	0.051	Access Point1	Laptop5	POP3
	0.051	B2_F1_Switch	B2_CoreRouter	POP3
	0.052	B2_CoreRouter	B2_EdgeRouter	POP3
	0.053	B2_EdgeRouter	ISP_Router	POP3
	0.054	ISP_Router	B1_EdgeRouter	POP3
	0.055	B1_EdgeRouter	B1_CoreRouter	POP3
	0.056	B1_CoreRouter	B1_F3_Switch	POP3
	0.057	B1_F3_Switch	MAIL	POP3
	0.058	MAIL	B1_F3_Switch	POP3
	0.059	B1_F3_Switch	B1_CoreRouter	POP3
	0.060	B1_CoreRouter	B1_EdgeRouter	POP3
	0.061	B1_EdgeRouter	ISP_Router	POP3
	0.062	ISP_Router	B2_EdgeRouter	POP3
	0.063	B2_EdgeRouter	B2_CoreRouter	POP3
	0.064	B2_CoreRouter	B2_F1_Switch	POP3
	0.065	B2_F1_Switch	Access Point1	POP3
	0.066	Access Point1	Tablet PC4	POP3
	0.066	Access Point1	Laptop3	POP3
	0.066	Access Point1	Laptop4	POP3
	0.066	Access Point1	Laptop5	POP3
	0.066	Access Point1	Laptop6	POP3
	0.066	Access Point1	Laptop7	POP3
	0.066	Access Point1	Tablet PC0	POP3
	0.066	Access Point1	Tablet PC2	POP3
	0.066	Access Point1	Tablet PC1	POP3
	0.066	Access Point1	Tablet PC3	POP3

Figure 3.1.6: Event List POP3

3.2 Scenario 2 – FTP File Transfer from B2-F2 to B1-F3

Network Functionality

In this scenario, a wired workstation located in Branch 2, Facility 2 initiates an FTP session to the centralized FTP Server in Branch 1, Facility 3 to upload a configuration file. The process involves establishing a control connection followed by a data connection to ensure reliable file transfer across the MAN.

1) Application & PDU Parameters:

The "Command Prompt" application was launched on the source workstation. The user executed the ftp 192.168.30.20 command. The PDU parameters were automatically configured by the system to target TCP Port 21 for control signaling. After successful authentication, the PDU parameters shifted to TCP Port 20 for the actual data payload transfer.

The PDU Information window reveals how the file data is segmented and addressed as it leaves Facility 2:

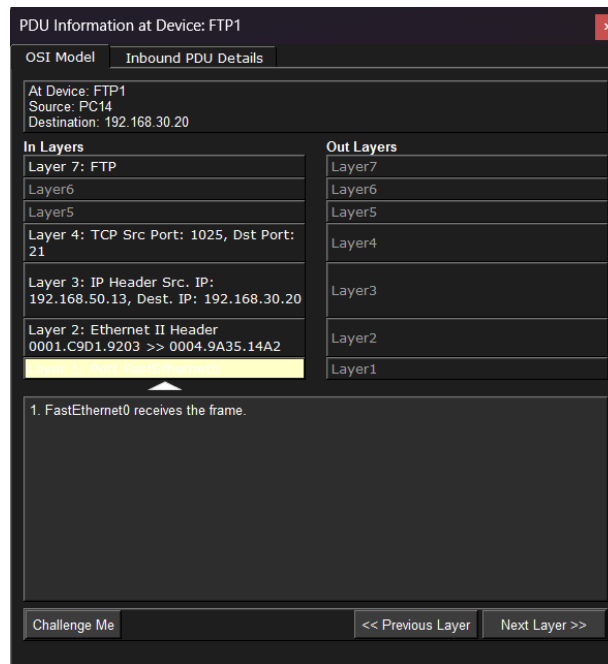


Figure 3.2.1: OSI Model at Device: FTP1(FTP)

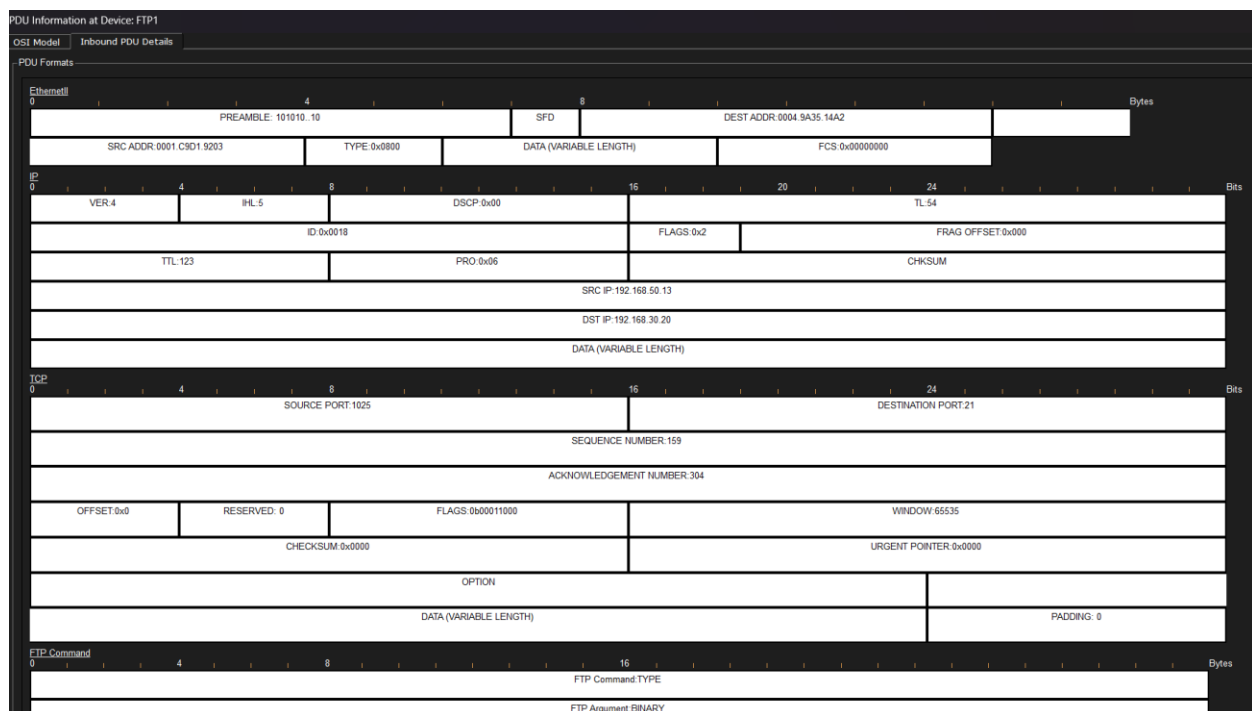


Figure 3.2.2: PDU Information at Device: FTP1(FTP)

2) Relevant Events Lists:

The table below illustrates the FTP PDU's path across the ISP backbone via OSPF-learned routes.

Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC14	FTP
	0.001	PC14	B2_F2_Switch	FTP
	0.002	B2_F2_Switch	B2_CoreRouter	FTP
	0.003	B2_CoreRouter	B2_EdgeRouter	FTP
	0.004	B2_EdgeRouter	ISP_Router	FTP
	0.005	ISP_Router	B1_EdgeRouter	FTP
	0.006	B1_EdgeRouter	B1_CoreRouter	FTP
	0.007	B1_CoreRouter	B1_F3_Switch	FTP
	0.008	B1_F3_Switch	FTP1	FTP
	0.008	--	FTP1	FTP
	0.009	FTP1	B1_F3_Switch	FTP
	0.010	B1_F3_Switch	B1_CoreRouter	FTP
	0.011	B1_CoreRouter	B1_EdgeRouter	FTP
	0.012	B1_EdgeRouter	ISP_Router	FTP
	0.013	ISP_Router	B2_EdgeRouter	FTP
	0.014	B2_EdgeRouter	B2_CoreRouter	FTP
	0.015	B2_CoreRouter	B2_F2_Switch	FTP
	0.016	B2_F2_Switch	PC14	FTP
	0.016	--	PC14	FTP

Figure 3.2.3: Event List FTP

3.3 Scenario 3 – VoIP Call Between Two Users in B1-F2

Network Functionality

This scenario demonstrates intra-facility Voice over IP (VoIP) communication within Branch 1, Facility 2. Instead of physical IP phones, employees utilize the Cisco IP Communicator softphone software installed on their workstations. The scenario highlights the call setup signaling process managed by the CME (Call Manager Express) on the Core Router, followed by the actual voice payload transmission between the two computers.

1) Application & PDU Parameters:

The "Cisco IP Communicator" application, located under the Desktop tab of the source workstation, was launched. The target extension number was dialed using the software's graphical interface. This action automatically generated SCCP (Skinny Client Control Protocol) PDUs for call signaling targeting the Router. Once the call was answered by the receiving workstation, the system generated continuous RTP (Real-time Transport Protocol) PDUs encapsulating the digitized voice data.

The PDU Information window reveals the distinct encapsulation method used for real-time, delay-sensitive voice traffic during the active call phase:

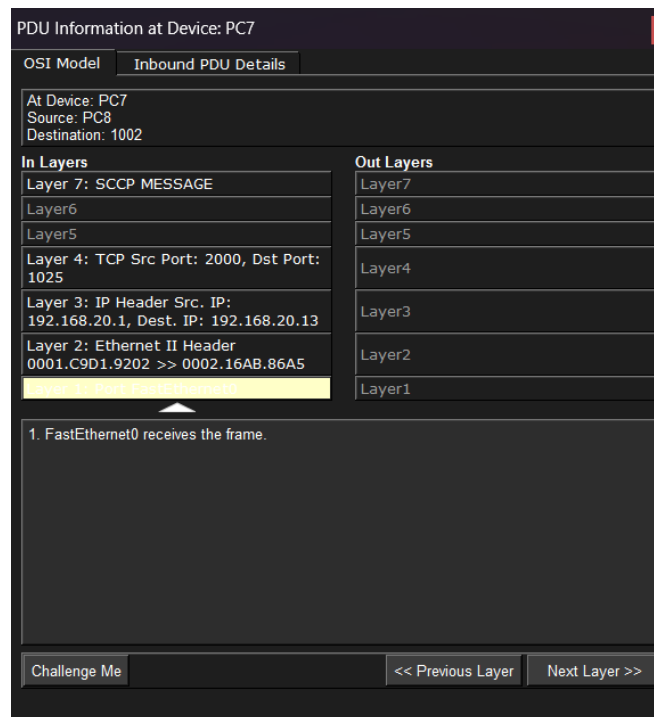


Figure 3.3.1: OSI Model at Device: PC7(SCCP)

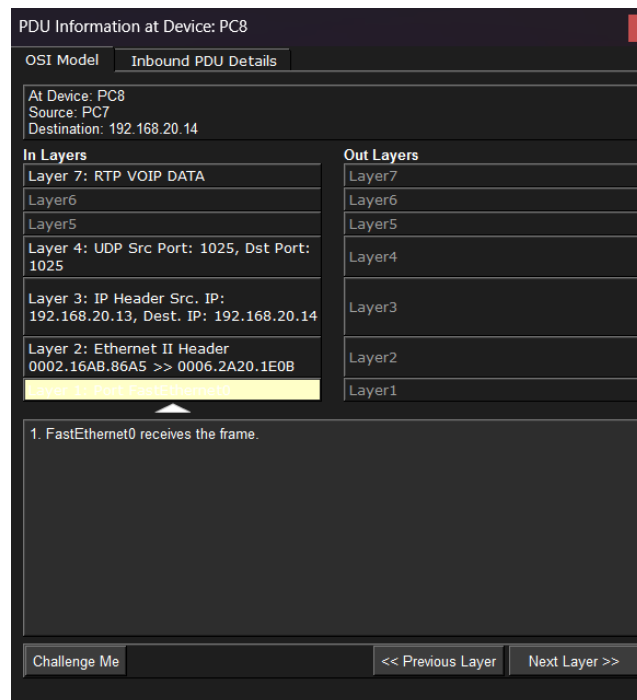


Figure 3.3.2: OSI Model at Device: PC8(RTP)

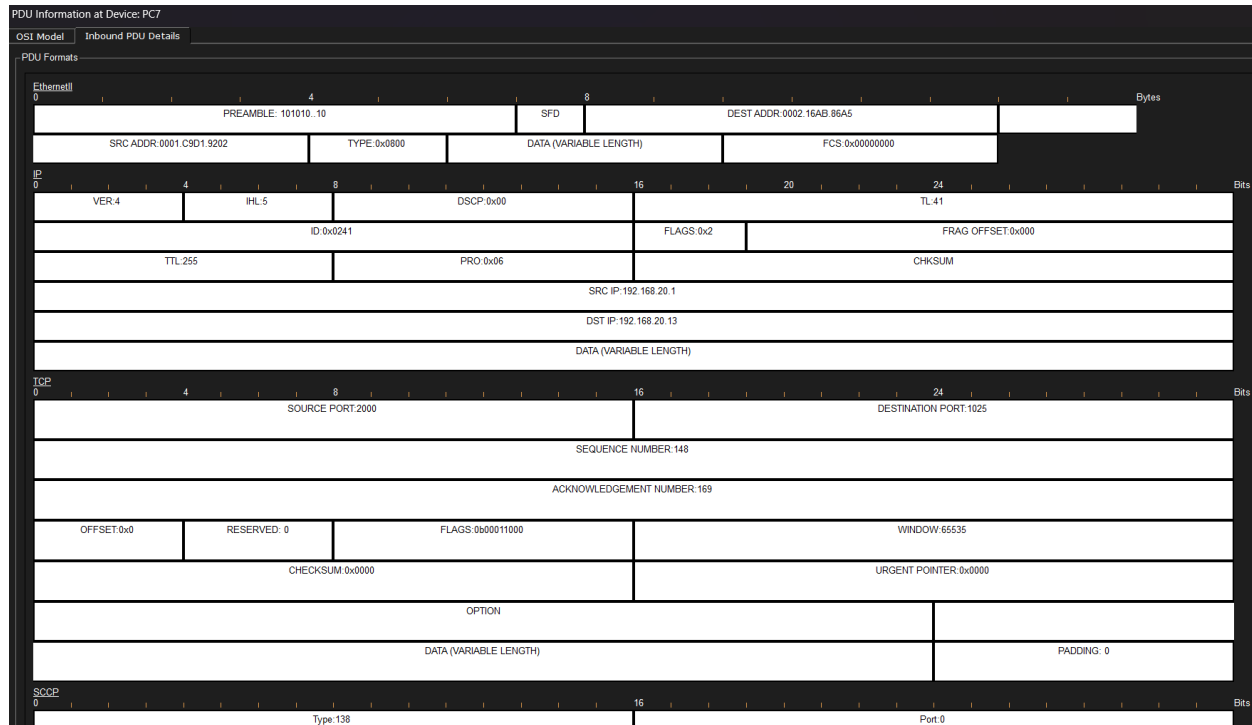


Figure 3.3.3: PDU Information at Device: PC7(SCCP)

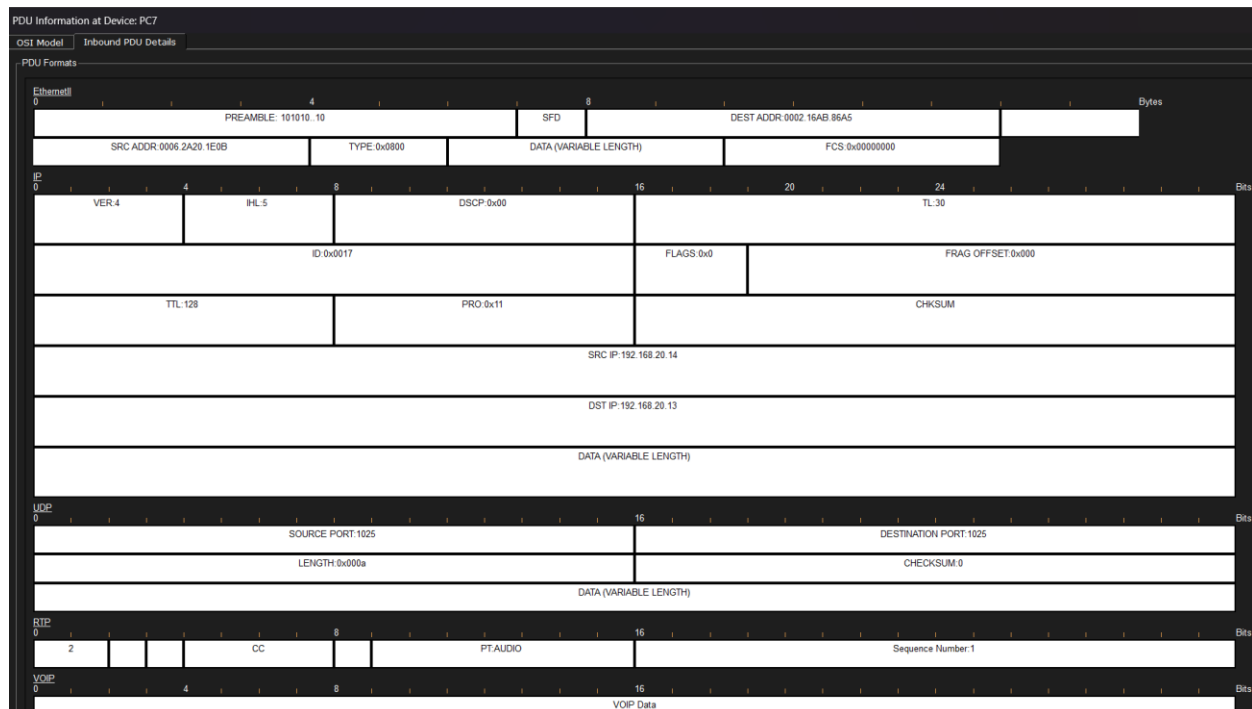


Figure 3.3.4: PDU Information at Device: PC8(RTP)

2) Relevant Events Lists:

The table below traces a single RTP voice packet after the call signaling has been established between the two workstations.

Vis.	Time(sec)	Last Device	At Device	Type
	9.126	PC8	B1_F2_Switch	SCCP
	9.126	B1_F2_Switch	B1_CoreRouter	SCCP
	9.127	B1_F2_Switch	B1_CoreRouter	SCCP
	9.127	B1_CoreRouter	B1_F2_Switch	SCCP
	9.128	B1_CoreRouter	B1_F2_Switch	SCCP
	9.128	B1_F2_Switch	PC8	SCCP
	9.128	--	PC8	RTP
	9.129	B1_F2_Switch	PC7	SCCP
	9.129	--	PC7	RTP
	9.132	--	PC8	RTP
	9.133	PC8	B1_F2_Switch	RTP
	9.133	--	PC7	RTP
	9.134	PC7	B1_F2_Switch	RTP
	9.134	B1_F2_Switch	PC7	RTP
	9.135	B1_F2_Switch	PC8	RTP
	24.129	--	PC8	RTP
	24.129	--	PC7	RTP
	24.130	PC8	B1_F2_Switch	RTP
	24.130	PC7	B1_F2_Switch	RTP
	24.131	B1_F2_Switch	PC7	RTP
	24.131	B1_F2_Switch	PC8	RTP
	30.723	--	B1_CoreRouter	SCCP
	30.724	B1_CoreRouter	B1_F2_Switch	SCCP
	30.724	--	B1_CoreRouter	SCCP
	30.725	B1_CoreRouter	B1_F2_Switch	SCCP
	30.725	B1_F2_Switch	PC7	SCCP
	30.726	B1_F2_Switch	PC8	SCCP
	34.281	--	PC7	SCCP
	34.282	PC7	B1_F2_Switch	SCCP
	34.283	B1_F2_Switch	B1_CoreRouter	SCCP

Figure 3.3.5: Event List VoIP

3.4 Scenario 4 – Email from B1-F2 to B2-F2

Network Functionality

This scenario tests the internal and inter-branch mail delivery system. A user in Branch 1, Facility 2 sends an email addressed to a colleague located in Branch 2, Facility 2. From a network topology perspective, email does not travel peer-to-peer. Instead, the sender's workstation transmits the email to the centralized Mail Server (in B1-F3) using SMTP.

1) Application & PDU Parameters:

The "Email" application within the Desktop GUI of the source PC (B1-F2) was utilized. The user populated the "To:" field with the recipient's configured email address, entered a subject and body, and clicked 'Send'. This action generated an SMTP (Simple Mail Transfer Protocol) PDU targeting the Mail Server on TCP Port 25.

The PDU Information window captures the packet as it leaves the sender's PC and route to the Mail Server:

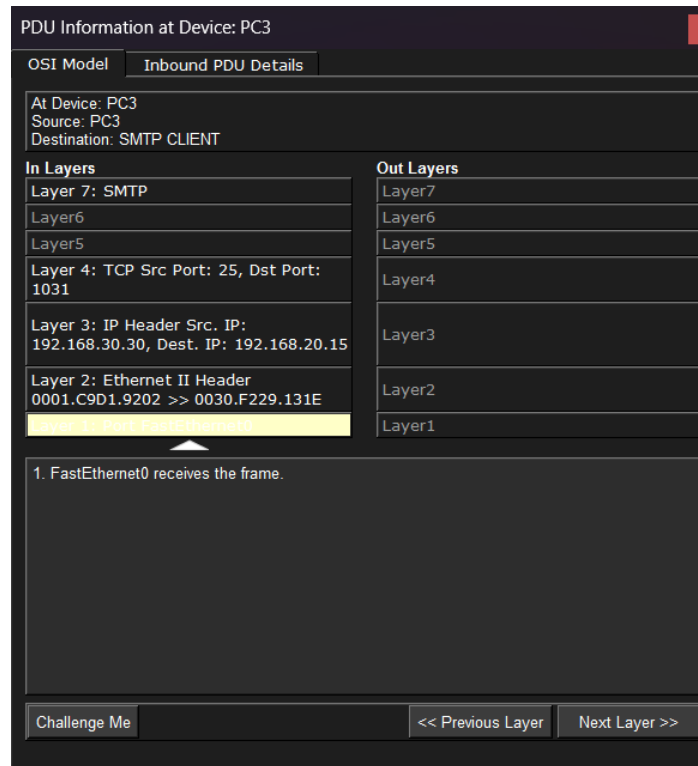


Figure 3.4.1: OSI Model at Device: PC3(SMTP)

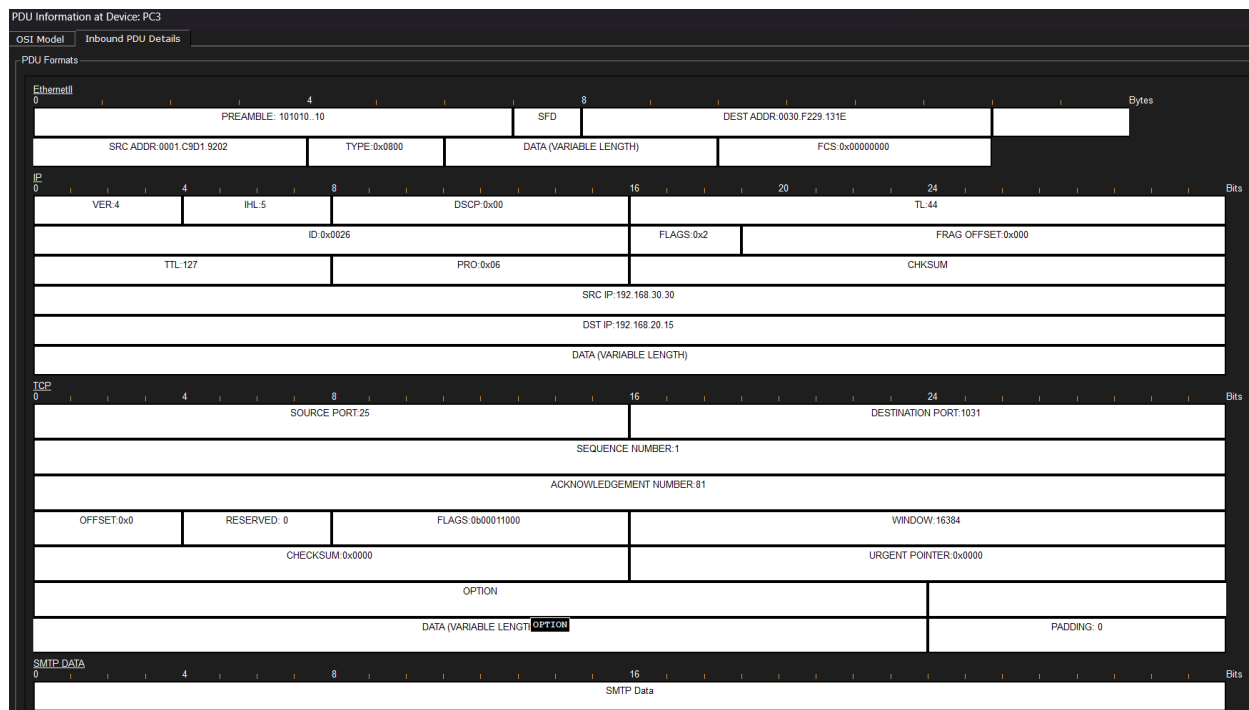
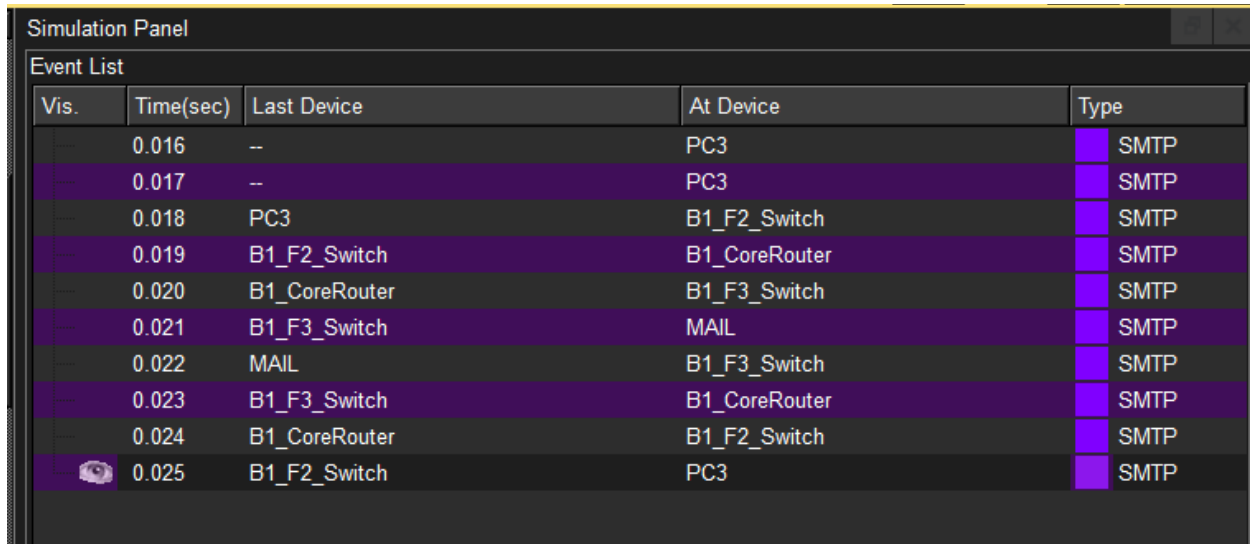


Figure 3.4.2: PDU Information at Device: PC3(SMTP)

2) Relevant Events Lists:

The following table traces the outbound SMTP routing path. Since the source (B1-F2) and the Mail Server (B1-F3) are in the same branch, the packet is routed internally by the Core Router without crossing the ISP backbone.

The image shows a screenshot of a 'Simulation Panel' with an 'Event List' table. The table has five columns: 'Vis.', 'Time(sec)', 'Last Device', 'At Device', and 'Type'. It contains 10 rows of data representing the steps of an SMTP routing process. The 'Type' column for all rows is 'SMTP'. The 'Last Device' and 'At Device' columns show the sequence of devices the packet passes through: PC3, B1_F2_Switch, B1_CoreRouter, B1_F3_Switch, MAIL, and back to B1_CoreRouter and B1_F2_Switch.

Vis.	Time(sec)	Last Device	At Device	Type
	0.016	--	PC3	SMTP
	0.017	--	PC3	SMTP
	0.018	PC3	B1_F2_Switch	SMTP
	0.019	B1_F2_Switch	B1_CoreRouter	SMTP
	0.020	B1_CoreRouter	B1_F3_Switch	SMTP
	0.021	B1_F3_Switch	MAIL	SMTP
	0.022	MAIL	B1_F3_Switch	SMTP
	0.023	B1_F3_Switch	B1_CoreRouter	SMTP
	0.024	B1_CoreRouter	B1_F2_Switch	SMTP
	0.025	B1_F2_Switch	PC3	SMTP

Figure 3.4.3: Event List SMTP

3.5 Scenario 5 – ICMP Ping from B2-F1 to Web Server in B1-F3

Network Functionality

This scenario tests the fundamental end-to-end IP connectivity and routing functionality across the entire Metropolitan Area Network. A user device located in Branch 2, Facility 1 pings the centralized Web Server located in Branch 1, Facility 3. This verifies that the OSPF dynamic routing protocol has successfully converged and that bidirectional communication is established across the ISP backbone.

1) Application & PDU Parameters:

The "Command Prompt" application on the source device's Desktop GUI was utilized. The command ping 192.168.30.10 was entered. This action directed the operating system to construct and transmit an ICMP (Internet Control Message Protocol).

Unlike HTTP or FTP, ICMP is a network-layer protocol. Therefore, the PDU Information window reveals that layers 4 and 7 are bypassed:

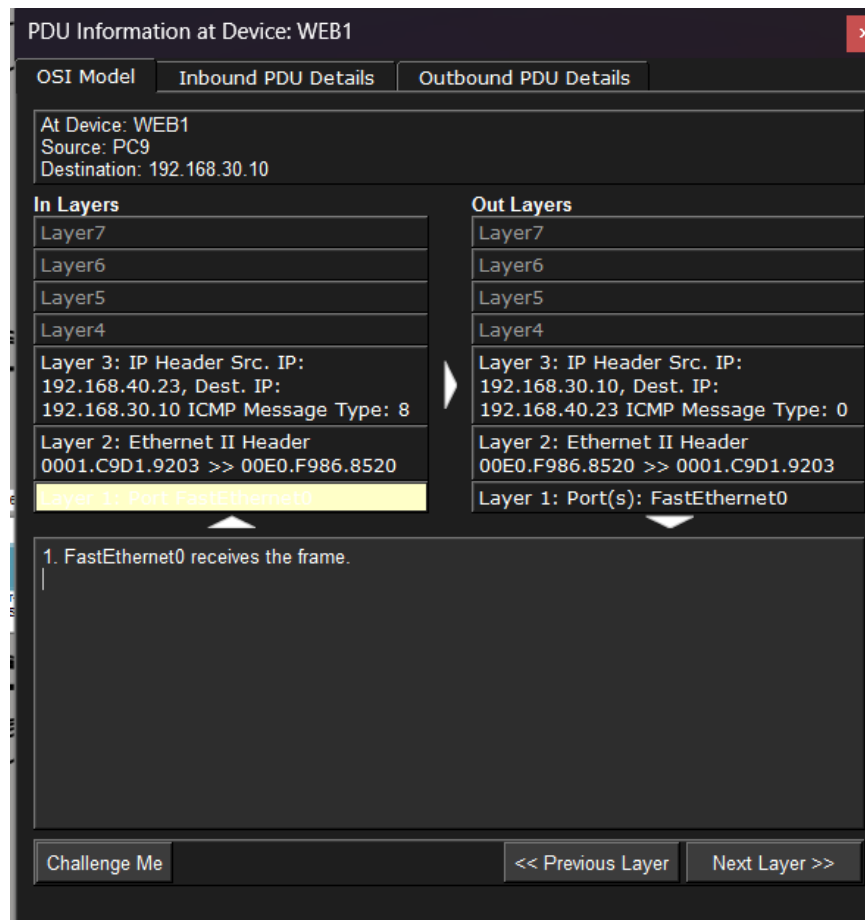


Figure 3.5.1: OSI Model at Device: WEB1(ICMP)

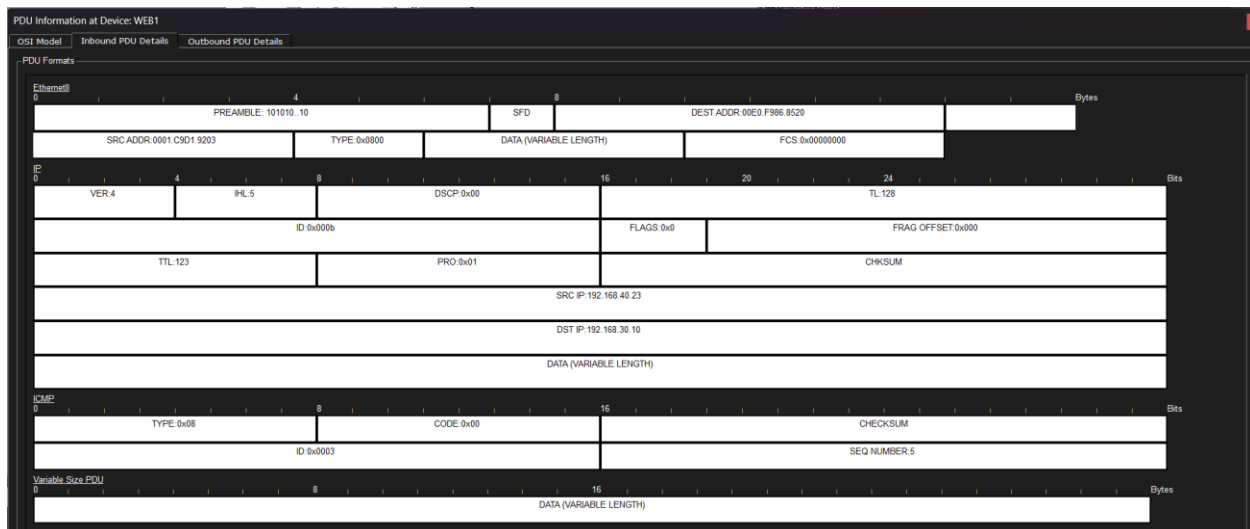
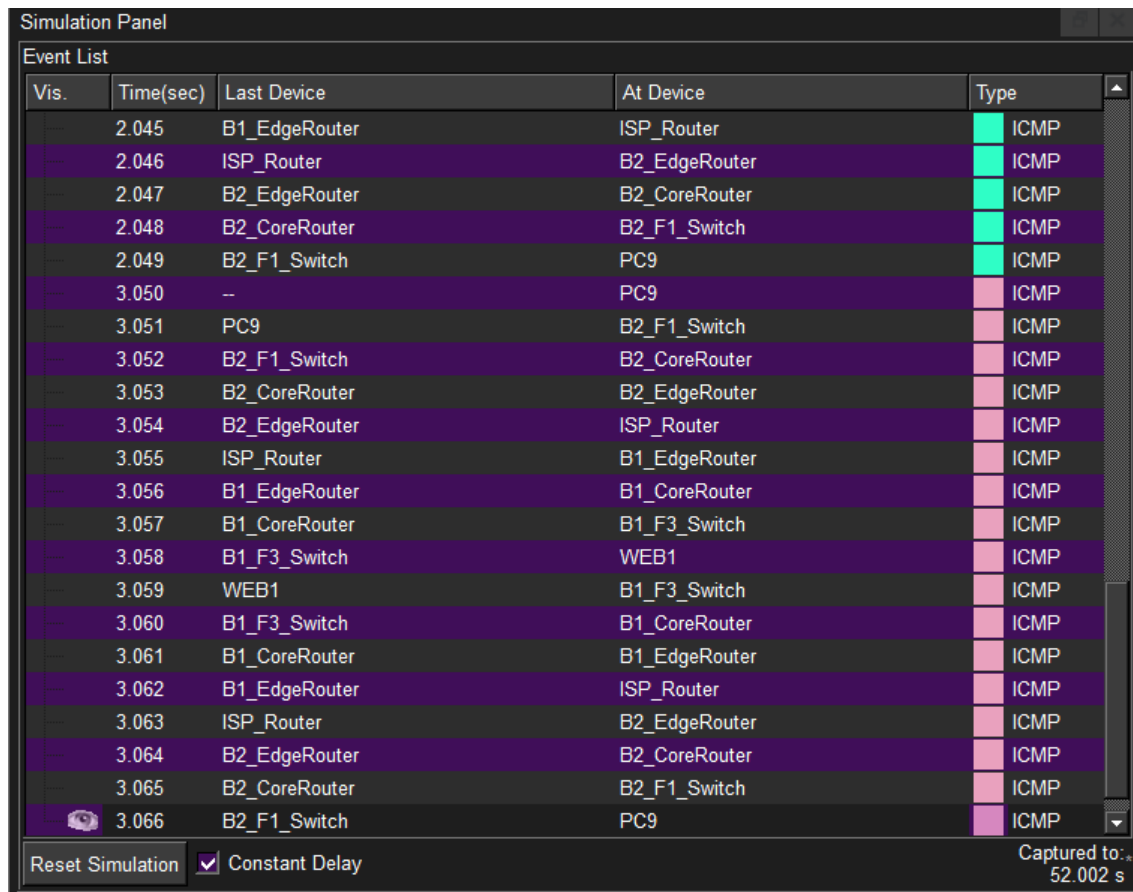


Figure 3.5.2: PDU Information at Device: WEB1(ICMP)

2) Relevant Events Lists:

The table below traces the ICMP Echo Request traveling to the Web Server, followed by the ICMP Echo Reply returning to the source.



The image shows a 'Simulation Panel' window with an 'Event List' table. The table has five columns: 'Vis.', 'Time(sec)', 'Last Device', 'At Device', and 'Type'. It contains 18 rows of events, all of which are ICMP. The first 9 rows (2.045 to 3.054) represent the outgoing ICMP Echo Request, and the next 9 rows (3.055 to 3.066) represent the incoming ICMP Echo Reply. The 'Type' column uses color coding: cyan for outgoing requests and pink for incoming replies. At the bottom of the panel, there is a 'Reset Simulation' button, a checked 'Constant Delay' checkbox, and a 'Captured to: 52.002 s' indicator.

Vis.	Time(sec)	Last Device	At Device	Type
	2.045	B1_EdgeRouter	ISP_Router	ICMP
	2.046	ISP_Router	B2_EdgeRouter	ICMP
	2.047	B2_EdgeRouter	B2_CoreRouter	ICMP
	2.048	B2_CoreRouter	B2_F1_Switch	ICMP
	2.049	B2_F1_Switch	PC9	ICMP
	3.050	--	PC9	ICMP
	3.051	PC9	B2_F1_Switch	ICMP
	3.052	B2_F1_Switch	B2_CoreRouter	ICMP
	3.053	B2_CoreRouter	B2_EdgeRouter	ICMP
	3.054	B2_EdgeRouter	ISP_Router	ICMP
	3.055	ISP_Router	B1_EdgeRouter	ICMP
	3.056	B1_EdgeRouter	B1_CoreRouter	ICMP
	3.057	B1_CoreRouter	B1_F3_Switch	ICMP
	3.058	B1_F3_Switch	WEB1	ICMP
	3.059	WEB1	B1_F3_Switch	ICMP
	3.060	B1_F3_Switch	B1_CoreRouter	ICMP
	3.061	B1_CoreRouter	B1_EdgeRouter	ICMP
	3.062	B1_EdgeRouter	ISP_Router	ICMP
	3.063	ISP_Router	B2_EdgeRouter	ICMP
	3.064	B2_EdgeRouter	B2_CoreRouter	ICMP
	3.065	B2_CoreRouter	B2_F1_Switch	ICMP
	3.066	B2_F1_Switch	PC9	ICMP

Figure 3.5.3: Event List ICMP

3.6 Scenario 6 – Laptop Email from B1-F1 to Friend in B2-F1

Network Functionality

This scenario evaluates both wireless connectivity and the internal routing of application-layer services. A user on a wireless laptop in Branch 1, Facility 1 sends an email to a friend located in Branch 2, Facility 1. Similar to the wired email scenario, the transmission is not peer-to-peer. The wireless laptop transmits the email to the centralized Mail Server located in B1-F3. The physical layer differs here, as the initial transmission occurs over

the B1_F1_Wifi SSID, secured by WPA2-PSK encryption, before entering the wired infrastructure.

1) Application & PDU Parameters:

The built-in "Email" application was opened on the B1-F1 wireless laptop's GUI. The user composed a message addressed to the B2-F1 user and clicked 'Send'. The operating system generated an SMTP (Simple Mail Transfer Protocol) PDU. The parameters were automatically set to target the IP address of the Mail Server using TCP Port 25.

The PDU Information window captures the packet as it leaves the wireless network interface card of the laptop:

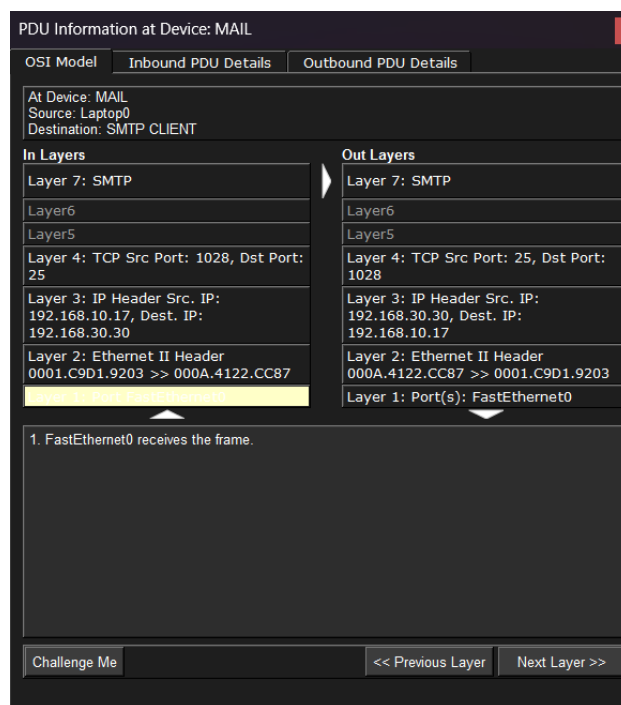


Figure 3.6.1: OSI Model at Device: MAIL(SMTP)

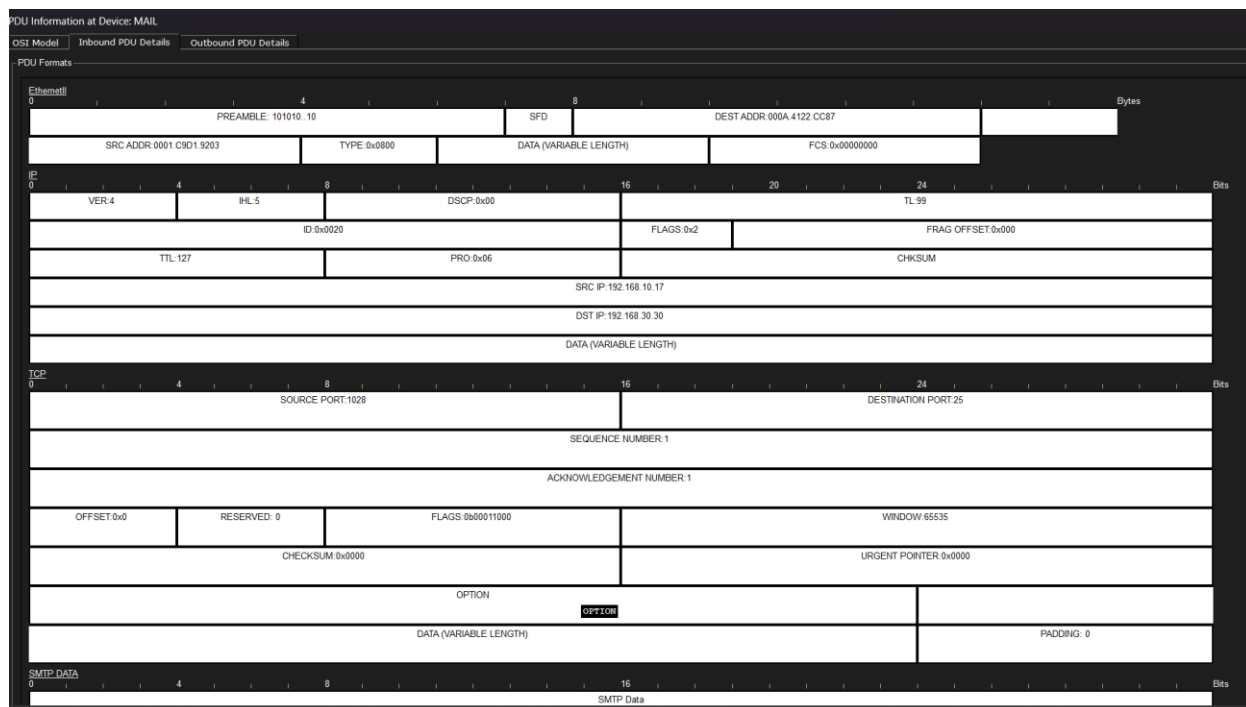


Figure 3.6.2: PDU Information at Device: MAIL(SMTP)

2) Relevant Events Lists:

The routing path below traces the email from the wireless laptop to the centralized Mail Server. Since the server is in the same branch, the packet does not cross the ISP router during the sending phase.

Vis.	Time(sec)	Last Device	At Device	Type
	0.027	Laptop0	Access Point0	SMTP
	0.028	Access Point0	B1_F1_Switch	SMTP
	0.029	B1_F1_Switch	B1_CoreRouter	SMTP
	0.030	B1_CoreRouter	B1_F3_Switch	SMTP
	0.031	B1_F3_Switch	MAIL	SMTP
	0.032	MAIL	B1_F3_Switch	SMTP
	0.032	--	Access Point0	SMTP
	0.033	Access Point0	Laptop0	SMTP
	0.033	Access Point0	Laptop1	SMTP
	0.033	Access Point0	Smartphone0	SMTP
	0.033	Access Point0	Smartphone1	SMTP
	0.033	Access Point0	Laptop2	SMTP
	0.033	Access Point0	Smartphone2	SMTP
	0.033	B1_F3_Switch	B1_CoreRouter	SMTP
	0.034	B1_CoreRouter	B1_F1_Switch	SMTP
	0.035	B1_F1_Switch	Access Point0	SMTP
	0.036	Access Point0	Laptop0	SMTP
	0.036	Access Point0	Laptop2	SMTP
	0.036	Access Point0	Laptop1	SMTP
	0.036	Access Point0	Smartphone0	SMTP
	0.036	Access Point0	Smartphone1	SMTP
	0.036	Access Point0	Smartphone2	SMTP

Figure 3.6.3: Event List SMTP

3.7 Scenario 7 – SSH Access from B2-F3 Mobile Device to B1_Core Router

Network Functionality

This scenario demonstrates secure remote management capabilities across the Metropolitan Area Network. A network administrator utilizing a mobile device in Branch 2, Facility 3 initiates a Secure Shell (SSH) connection to manage the B1_Core router located in Branch 1. *(Note: While the initial topology design aimed to SSH into the Web Server, standard Packet Tracer server nodes do not natively support SSH daemon simulation. Therefore, targeting the Core Router perfectly validates the end-to-end routing of encrypted management traffic across the WAN).* This proves that the infrastructure can be safely administered remotely without exposing credentials, unlike legacy Telnet.

1) Application & PDU Parameters:

The "SSH Client" (or Command Prompt) application within the Desktop GUI of the mobile device was utilized. The connection was initiated using the command `ssh -l [username] [B1_Core_Router_IP]`. This action generated an SSH PDU. The transport layer parameters were automatically configured to target the router's VTY lines on TCP Port 22.

The PDU Information window captures the packet as the secure handshake begins:

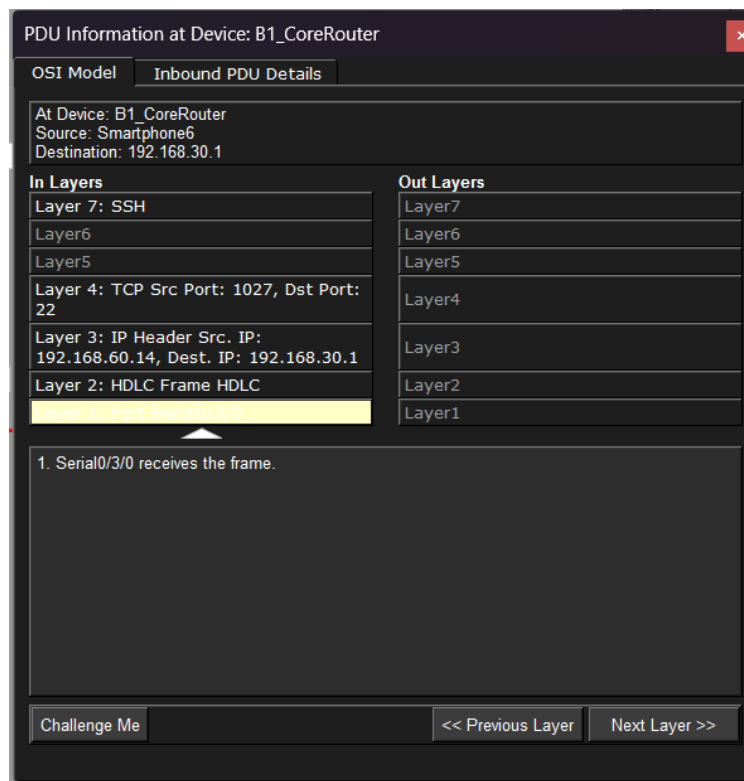


Figure 3.7.1: OSI Model at Device: B1_CoreRouter(SSH)

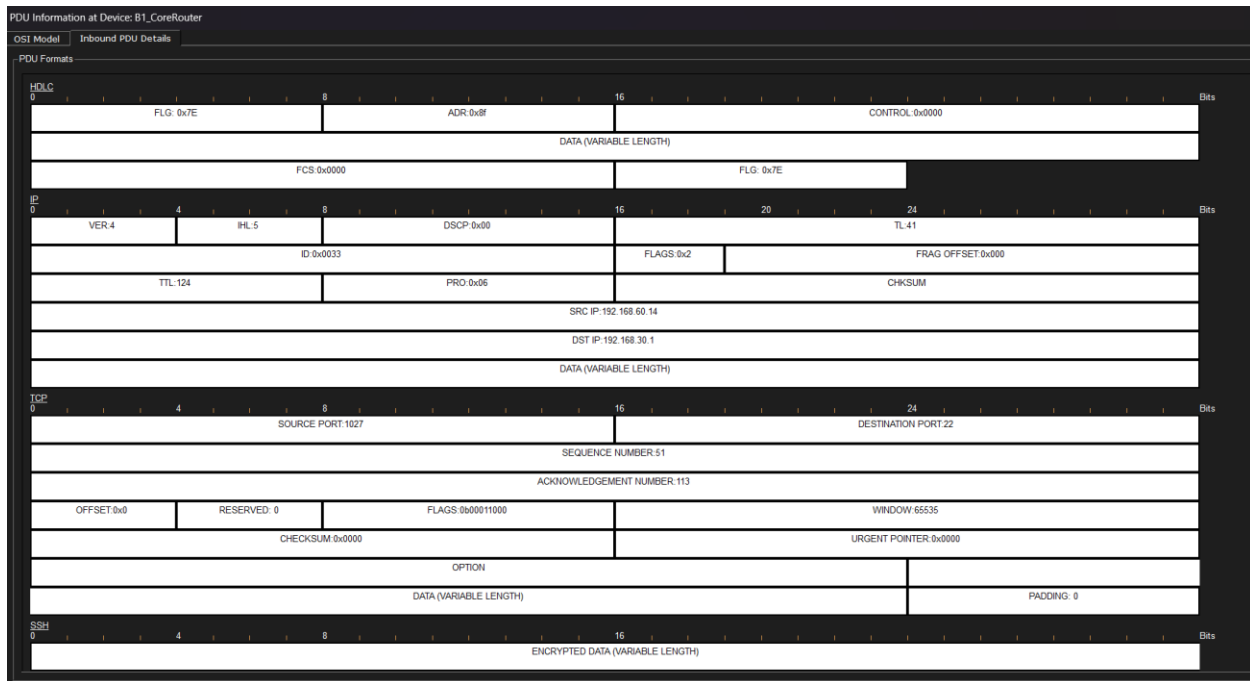


Figure 3.7.2: PDU Information at Device: B1_CoreRouter(SSH)

2) Relevant Events Lists:

The table below traces the SSH packet from the mobile client, across the wireless medium, through the OSPF-routed ISP backbone, and finally to the destination Core Router in Branch 1.

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	Smartphone6	SSH
	0.000	--	Smartphone6	SSH
	0.000	--	Smartphone6	SSH
	0.000	--	Smartphone6	SSH
	0.001	Smartphone6	Access Point3	SSH
	0.002	Access Point3	B2_F3_Switch	SSH
	0.003	B2_F3_Switch	B2_CoreRouter	SSH
	0.003	--	Access Point3	SSH
	0.004	Access Point3	Smartphone5	SSH
	0.004	Access Point3	Smartphone6	SSH
	0.004	B2_CoreRouter	B2_EdgeRouter	SSH
	0.005	B2_EdgeRouter	ISP_Router	SSH
	0.006	ISP_Router	B1_EdgeRouter	SSH
	0.007	B1_EdgeRouter	B1_CoreRouter	SSH
	0.007	--	B1_CoreRouter	SSH
	0.008	B1_CoreRouter	B1_EdgeRouter	SSH
	0.009	B1_EdgeRouter	ISP_Router	SSH
	0.010	ISP_Router	B2_EdgeRouter	SSH
	0.011	B2_EdgeRouter	B2_CoreRouter	SSH
	0.012	B2_CoreRouter	B2_F3_Switch	SSH
	0.013	B2_F3_Switch	Access Point3	SSH
	0.014	Access Point3	Smartphone6	SSH

Reset Simulation ☒ Constant Delay Captured to... 0.014 s

Figure 3.7.3: Event List SSH

3.8 Custom Scenario 1 – FTP File Download from B1-F3 to B1-F1

Network Functionality

This custom scenario is designed to test the persistence and accessibility of centralized services by retrieving the exact configuration file that was uploaded in Scenario 2. A user located in Branch 1, Facility 1 initiates an FTP session to the centralized FTP Server in Branch 1, Facility 3 to download the file.

1) Application & PDU Parameters:

The "Command Prompt" application was launched on the source PC in B1-F1. The user established the connection using the [ftp 192.168.30.23](ftp://192.168.30.23) command. After authenticating, the get app.html command was issued to initiate the download. The PDU parameters utilized TCP Port 21 for the control channel (authenticating and sending the 'get' command) and TCP Port 20 for receiving the actual data payload from the server.

The PDU Information window captures the packet during the active download phase:

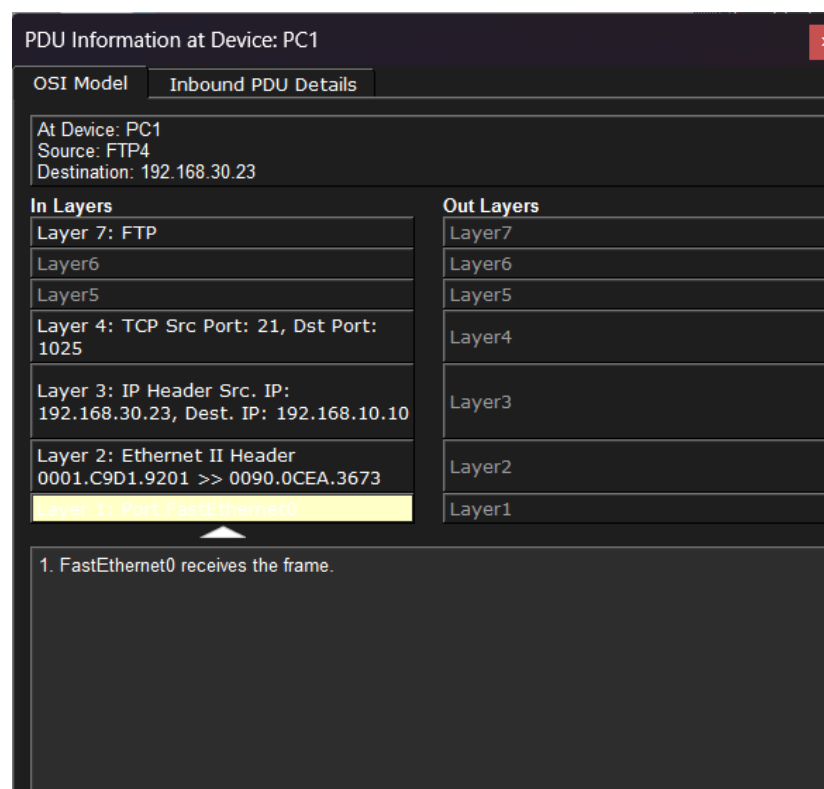
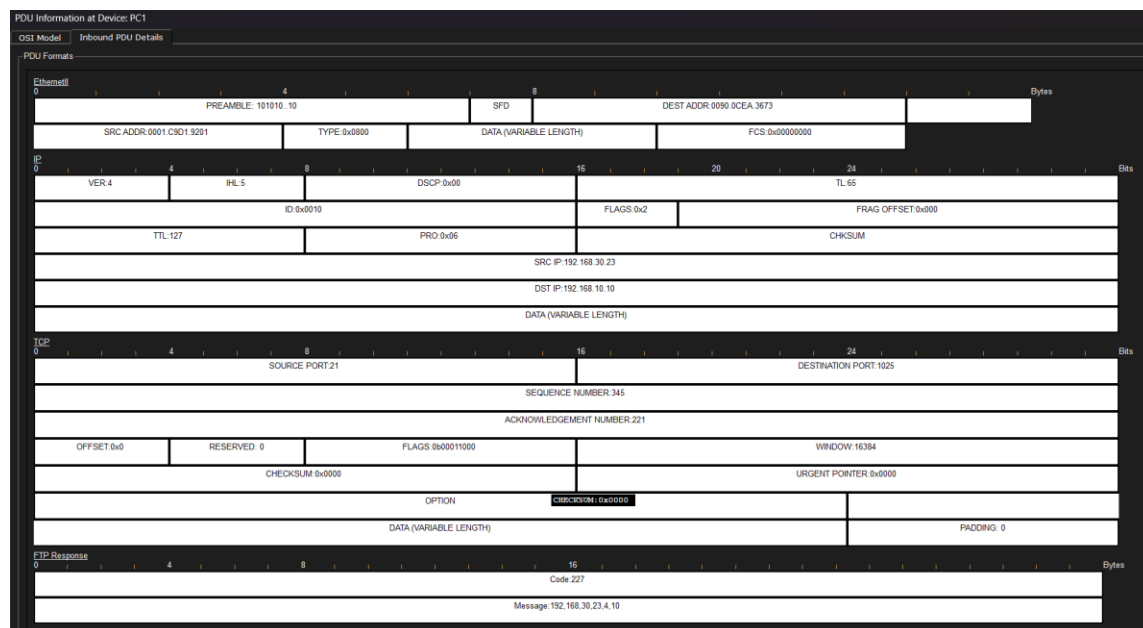


Figure 3.8.1: OSI Model at Device: PC1(FTP)



2) Relevant Events Lists:

The table below illustrates the FTP data packet's path. Notice that the packet does not travel to the Edge or ISP routers; it is routed internally by the Core Router..

Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
---	0.011	B1_CoreRouter	B1_F3_Switch	FTP
---	0.012	B1_F3_Switch	FTP4	FTP
---	0.012	--	FTP4	FTP
---	0.013	FTP4	B1_F3_Switch	FTP
---	0.014	B1_F3_Switch	B1_CoreRouter	FTP
---	0.015	B1_CoreRouter	B1_F1_Switch	FTP
---	0.016	B1_F1_Switch	PC1	FTP
---	0.016	--	PC1	FTP
---	0.017	PC1	B1_F1_Switch	FTP
---	0.018	B1_F1_Switch	B1_CoreRouter	FTP
---	0.019	B1_CoreRouter	B1_F3_Switch	FTP
---	0.020	B1_F3_Switch	FTP4	FTP
---	0.020	--	FTP4	FTP
---	0.021	FTP4	B1_F3_Switch	FTP
---	0.022	B1_F3_Switch	B1_CoreRouter	FTP
---	0.023	B1_CoreRouter	B1_F1_Switch	FTP
---	0.024	B1_F1_Switch	PC1	FTP
---	0.036	--	FTP4	FTP
---	0.037	FTP4	B1_F3_Switch	FTP
---	0.038	B1_F3_Switch	B1_CoreRouter	FTP
---	0.039	B1_CoreRouter	B1_F1_Switch	FTP
	0.040	B1_F1_Switch	PC1	FTP

3.9 Custom Scenario – Route Tracing (Traceroute) from B2-F3 to B1-F3 Server

Network Functionality

This custom scenario maps the exact routing path a packet takes across the Metropolitan Area Network. A user in Branch 2, Facility 3 traces the route to the centralized Web Server in Branch 1, Facility 3. This test proves that OSPF is actively managing the routing tables across multiple hops (Core, Edge, and ISP routers). The mechanism relies on sending packets with an incrementally increasing Time-To-Live (TTL) value.

1) Application & PDU Parameters:

The "Command Prompt" application on the source PC was used. The command `tracert 192.168.30.19` was executed. This instructed the operating system to send a sequence of ICMP Echo Request PDUs. The critical parameter here is the IP header's TTL field, which starts at 1 for the first packet and increments by 1 for subsequent packets to discover each router along the path.

The PDU Information window for the very first tracing packet (TTL=1) highlights the network-layer routing mechanisms:

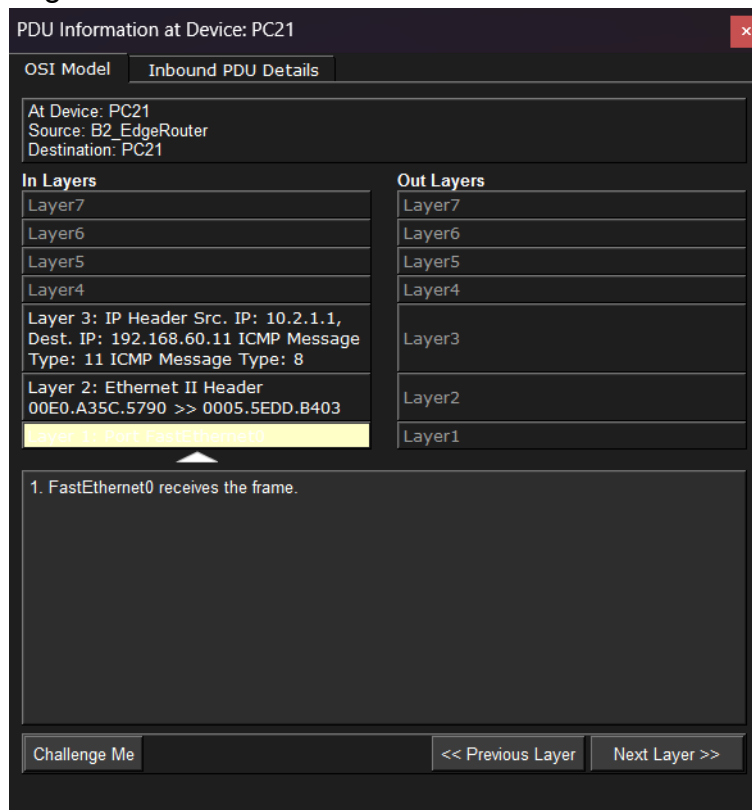


Figure 3.9.1: OSI Model at Device: PC21(ICMP)

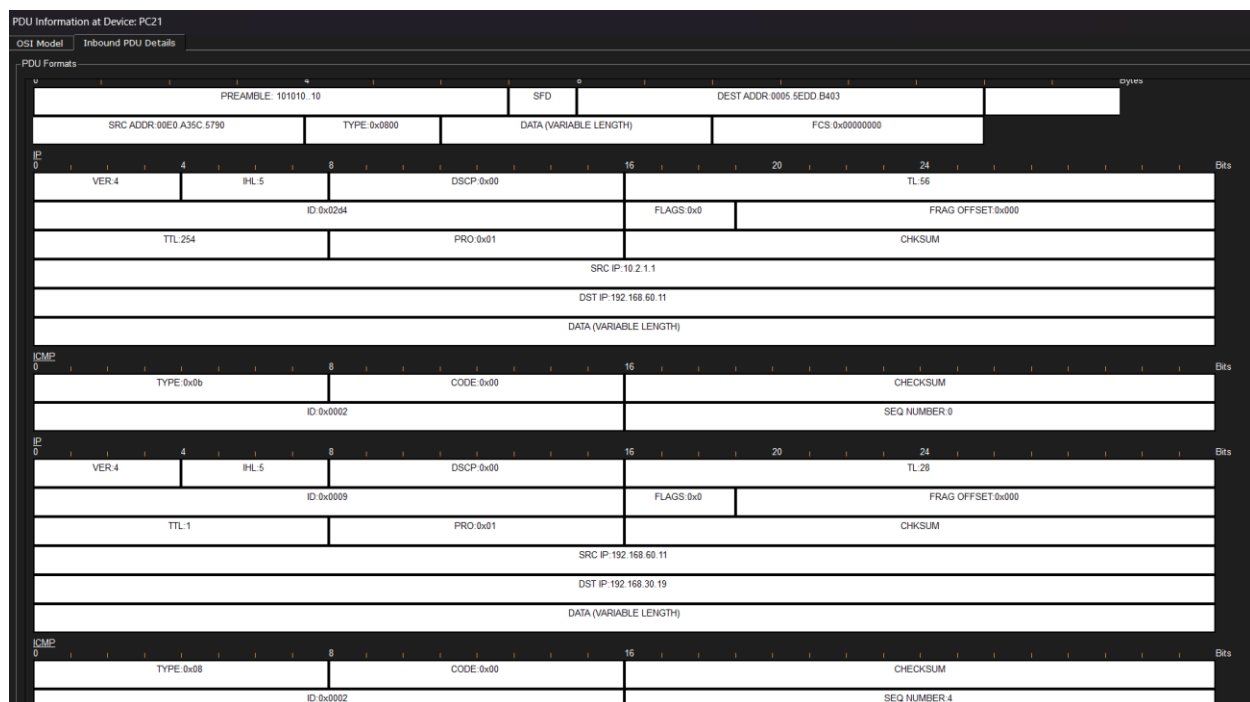


Figure 3.9.2: PDU Information at Device: PC21(ICMP)

2) Relevant Events Lists:

The table below traces the first hop of the tracer process. When the packet reaches the first router, its TTL drops to 0, causing the router to drop the packet and return an ICMP "Time Exceeded" message, thereby revealing its identity to the client.

Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.751	ISP_Router	B2_EdgeRouter	ICMP
	0.752	B2_EdgeRouter	B2_CoreRouter	ICMP
	0.753	B2_CoreRouter	B2_F3_Switch	ICMP
	0.754	B2_F3_Switch	PC21	ICMP
	0.856	--	PC21	ICMP
	0.857	PC21	B2_F3_Switch	ICMP
	0.858	B2_F3_Switch	B2_CoreRouter	ICMP
	0.859	B2_CoreRouter	B2_EdgeRouter	ICMP
	0.860	B2_EdgeRouter	ISP_Router	ICMP
	0.860	--	ISP_Router	ICMP
	0.861	ISP_Router	B2_EdgeRouter	ICMP
	0.862	B2_EdgeRouter	B2_CoreRouter	ICMP
	0.863	B2_CoreRouter	B2_F3_Switch	ICMP
	0.864	B2_F3_Switch	PC21	ICMP
	0.965	--	PC21	ICMP
	0.966	PC21	B2_F3_Switch	ICMP
	0.967	B2_F3_Switch	B2_CoreRouter	ICMP
	0.968	B2_CoreRouter	B2_EdgeRouter	ICMP
	0.969	B2_EdgeRouter	ISP_Router	ICMP
	0.970	ISP_Router	B1_EdgeRouter	ICMP
	0.970	--	B1_EdgeRouter	ICMP
	0.971	B1_EdgeRouter	ISP_Router	ICMP

Figure 3.9.3: Event List ICMP

4. Conclusion

This project successfully demonstrated the design and simulation of a Metropolitan Area Network using Cisco Packet Tracer. The network architecture accommodates two branch offices with diverse user populations and service requirements, interconnected through an ISP. Key services including HTTP, FTP, SMTP/POP3, VoIP, DNS, DHCP, and SSH were configured and tested through nine simulation scenarios.

The simulation confirmed that proper IP addressing, routing configuration, and service setup are essential prerequisites for reliable cross-branch communication. VoIP and SSH scenarios, in particular, highlighted the need for careful protocol configuration and security considerations in MAN environments.

5. References

- [1] A. Maria, "Introduction to modeling and simulation," in *Proceedings of the 29th Conference on Winter Simulation*, Atlanta, GA, USA, 1997, pp. 7–13.
- [2] T. Issariyakul and E. Hossain, "Simulation of Computer Networks," in *Introduction to Network Simulator NS2*, New York, NY, USA: Springer Science+Business Media, 2009.
- [3] Cisco Systems, "Cisco Packet Tracer," Cisco Networking Academy, 2024. [Online]. Available: <https://www.netacad.com/courses/packet-tracer>. [Accessed: May 2026].

Appendix A: Network Topology Diagram

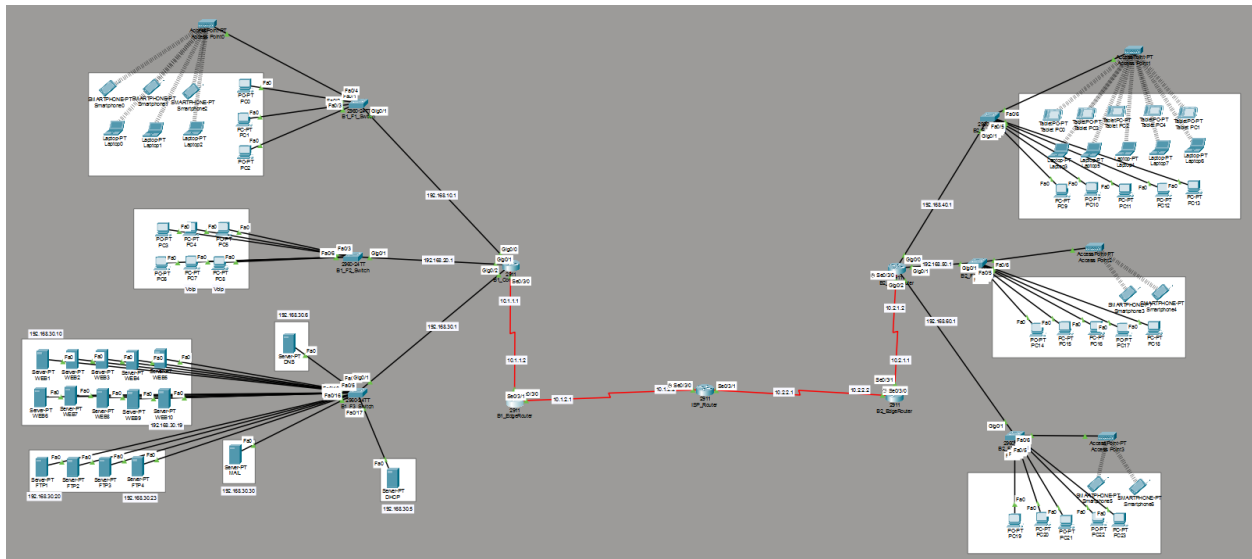


Figure A.1: Network Topology Diagram

Appendix B: Packet Tracer Screenshots – Scenario Results

Scenario 1 Screenshots

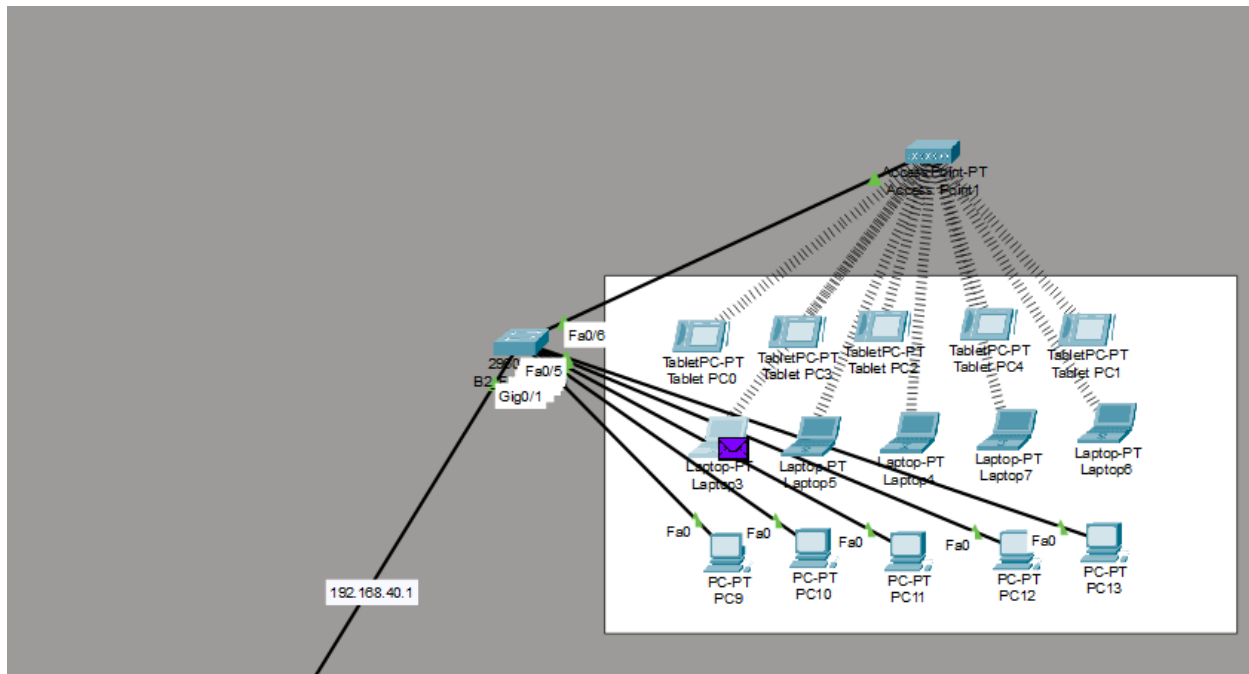


Figure B.1.1: Packet Transfer

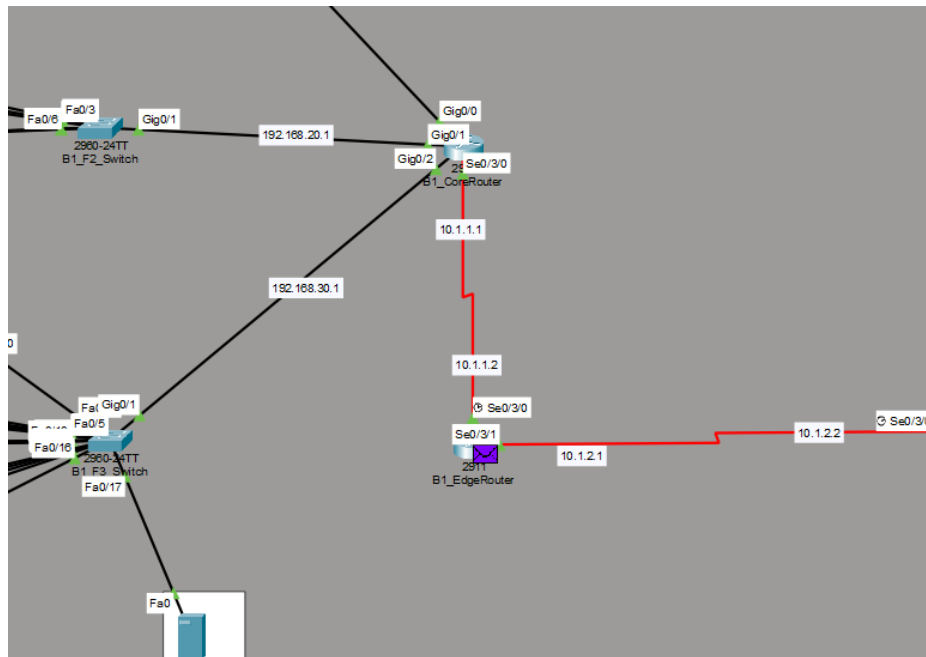


Figure B.1.2: Packet Transfer

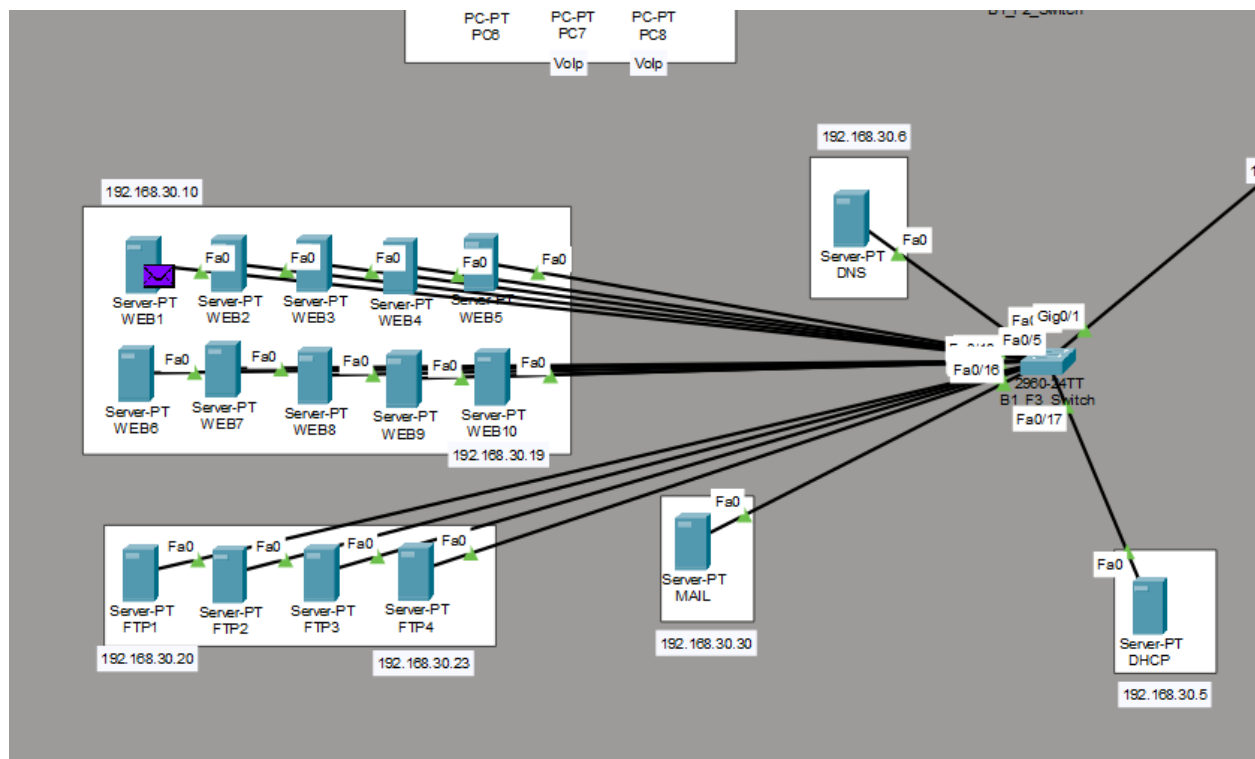


Figure B.1.3: Packet Transfer

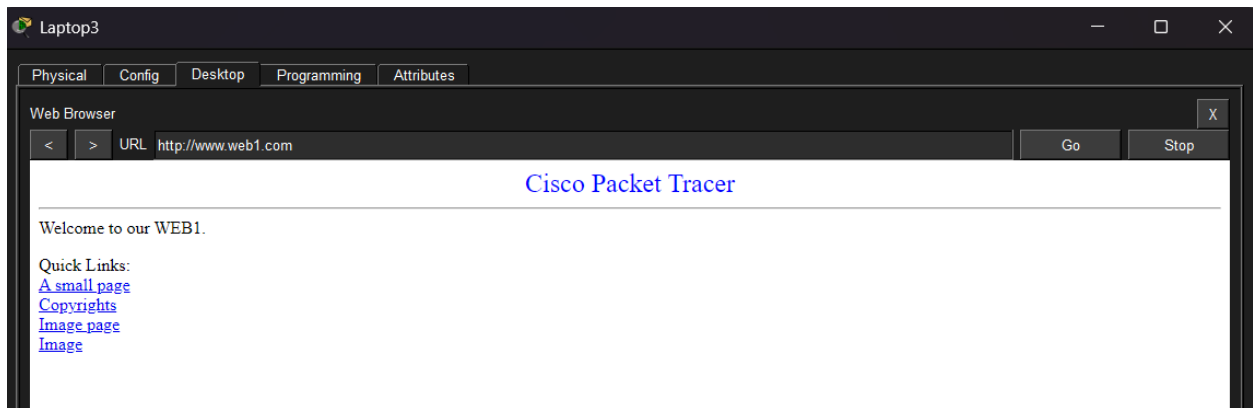


Figure B.1.4: Web Page

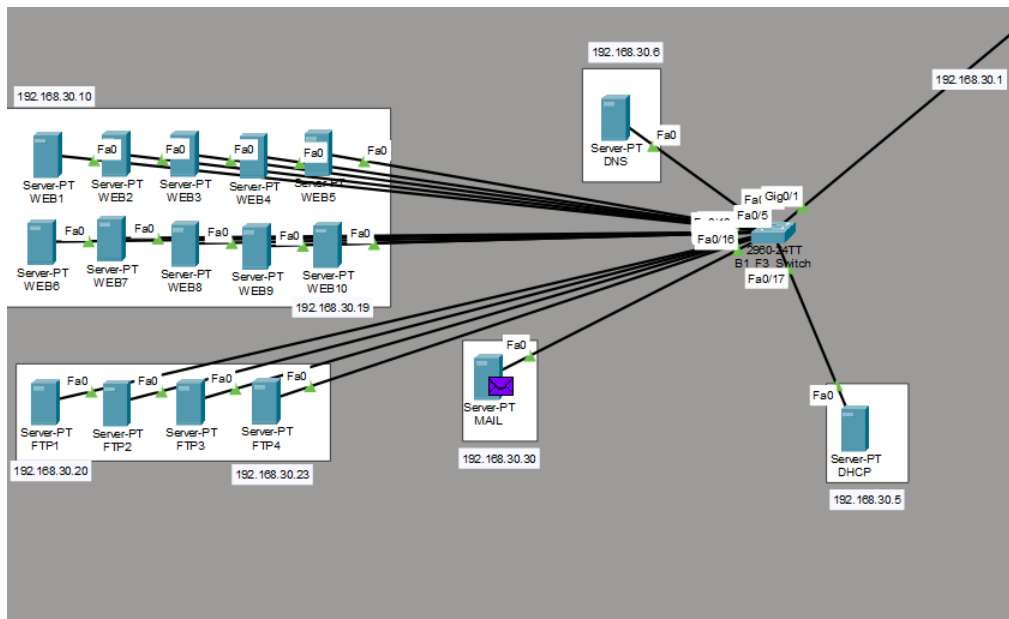


Figure B.1.5: Packet Transfer

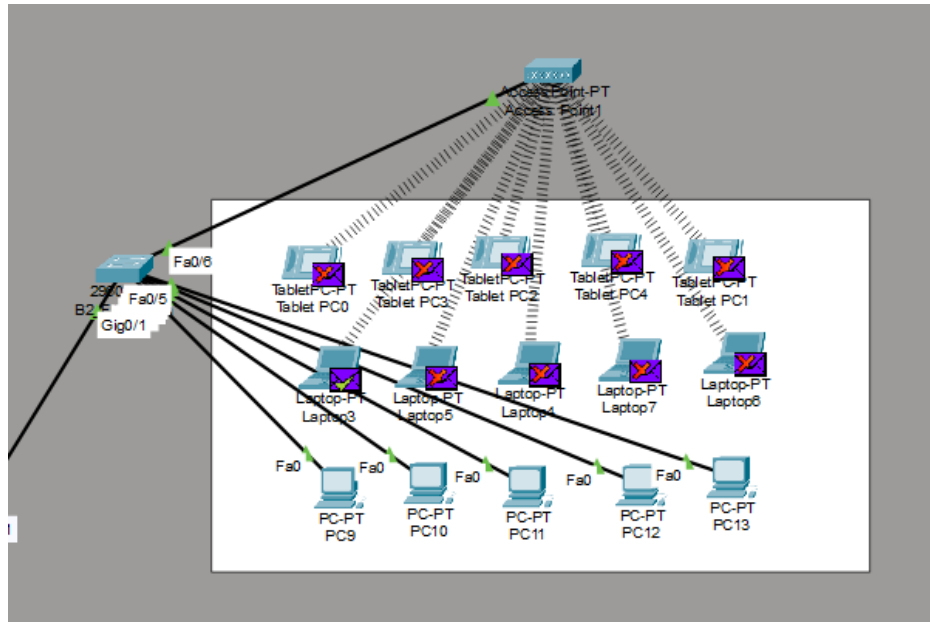


Figure B.1.6: Packet Transfer

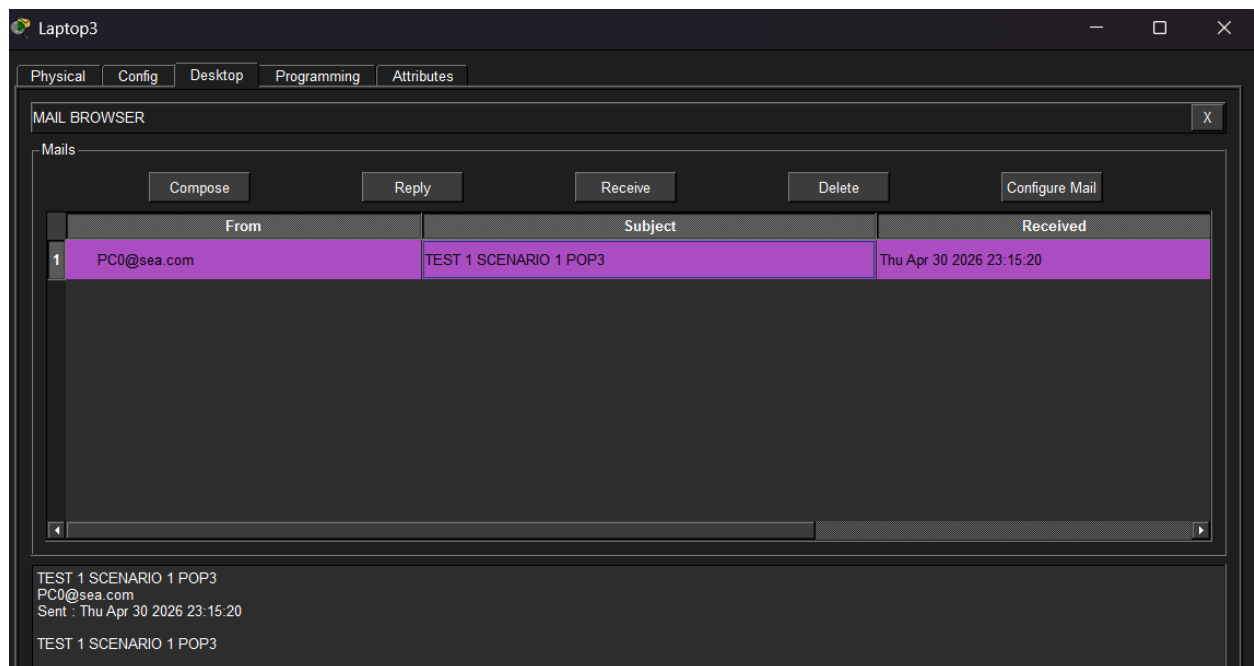


Figure B.1.7: Received Mail

Scenario 2 Screenshots

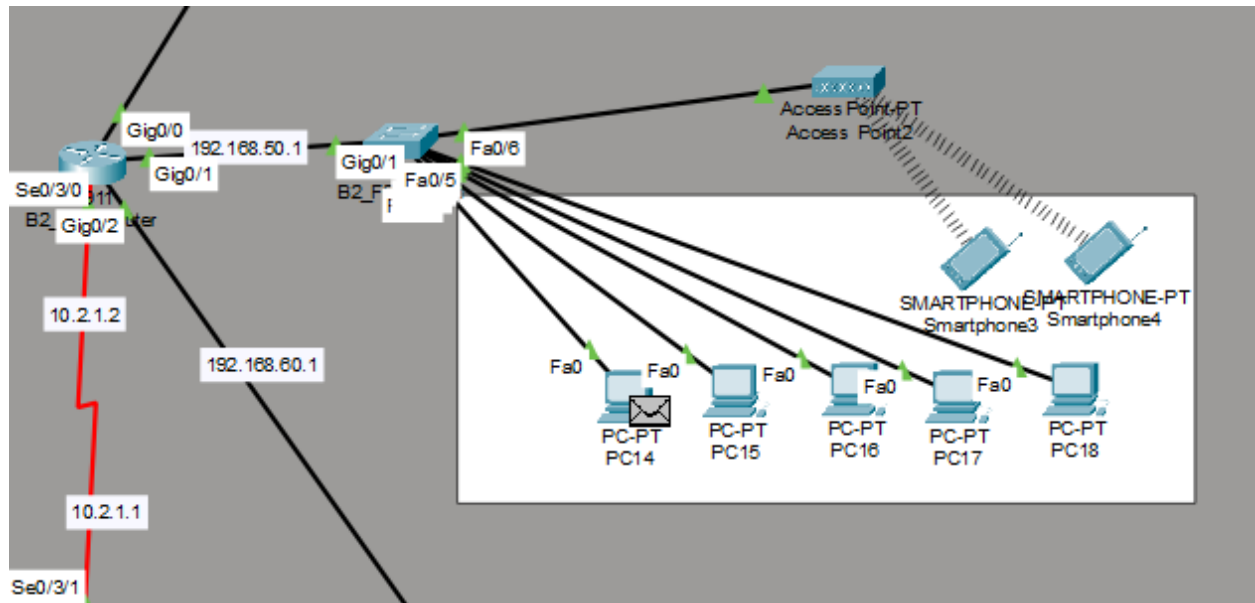


Figure B.2.1: Packet Transfer

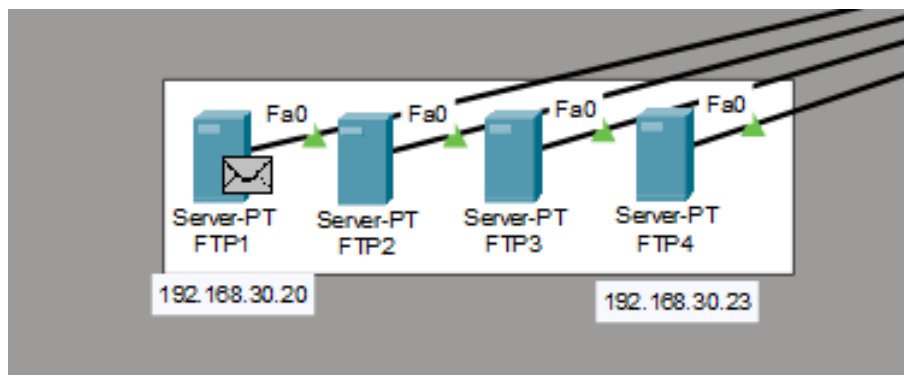
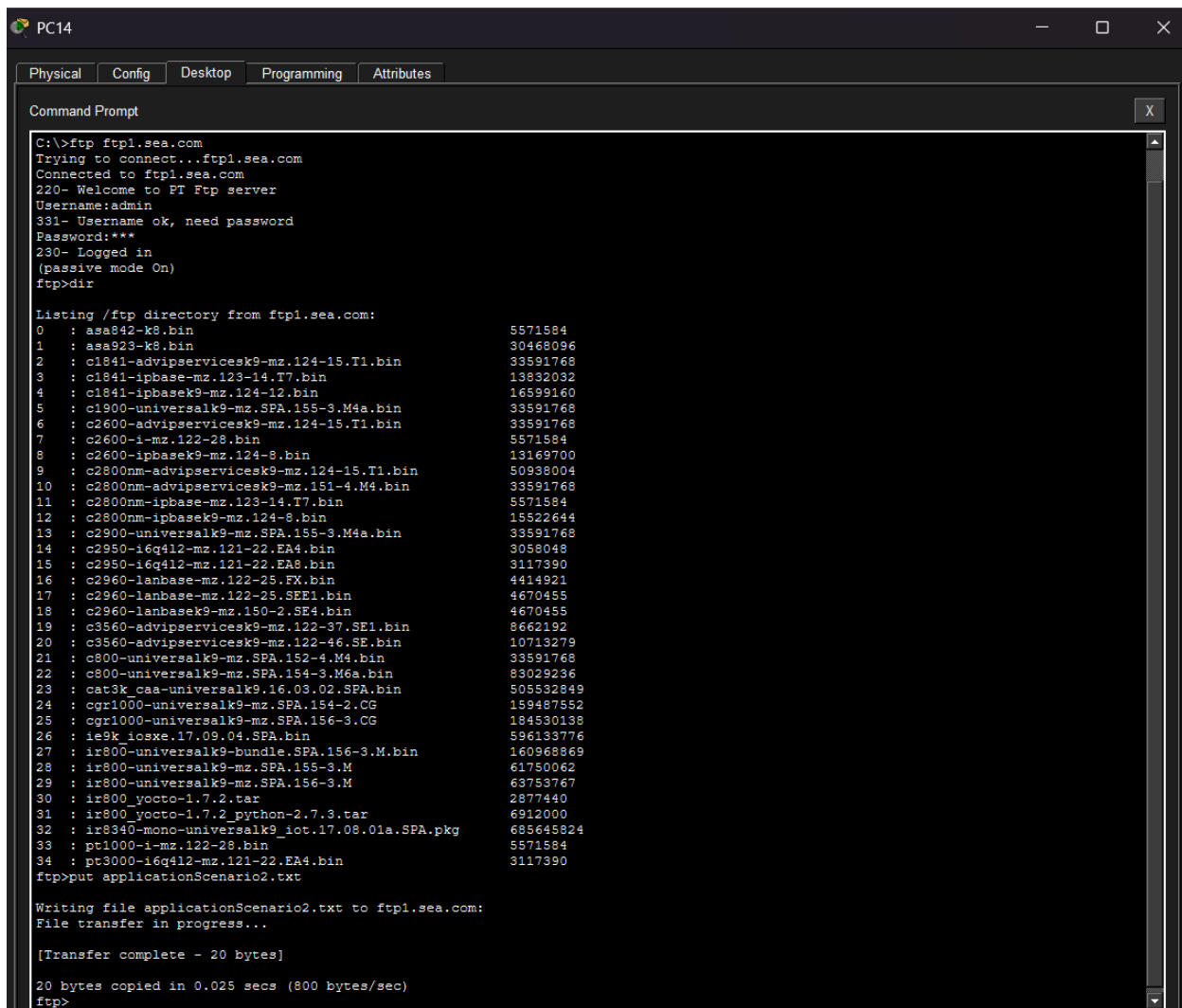


Figure B.2.2: Packet Transfer



The screenshot shows a PC14 window with a dark theme. At the top, there are tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Command Prompt' tab is active, displaying a series of FTP commands and their outputs. The commands include connecting to 'ftp1.sea.com', logging in as 'admin', listing the directory, and uploading a file. The directory listing shows a long list of files with their names and sizes. The upload command 'ftp>put applicationScenario2.txt' is followed by a confirmation message and a transfer progress indicator.

```
C:\>ftp ftp1.sea.com
Trying to connect...ftp1.sea.com
Connected to ftp1.sea.com
220- Welcome to FT Ftp server
Username:admin
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from ftp1.sea.com:
0 : asa842-k8.bin 5571584
1 : asa923-k8.bin 30468096
2 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
3 : c1841-ipbase-mz.123-14.T7.bin 13832032
4 : c1841-ipbasek9-mz.124-12.bin 16599160
5 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
6 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
7 : c2600-i-mz.122-28.bin 5571584
8 : c2600-ipbasek9-mz.124-8.bin 13169700
9 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
10 : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
11 : c2800nm-ipbase-mz.123-14.T7.bin 5571584
12 : c2800nm-ipbasek9-mz.124-8.bin 15522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
14 : c2950-16q412-mz.121-22.EA4.bin 3058048
15 : c2950-16q412-mz.121-22.EA8.bin 3117390
16 : c2960-lanbase-mz.122-25.FX.bin 4414921
17 : c2960-lanbase-mz.122-25.SEE1.bin 4670455
18 : c2960-lanbasek9-mz.150-2.SE4.bin 4670455
19 : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
20 : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
21 : c800-universalk9-mz.SPA.152-4.M4.bin 33591768
22 : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
23 : cat3k_gaa-universalk9.16.03.02.SPA.bin 505532849
24 : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
25 : cgr1000-universalk9-mz.SPA.156-3.CG 184530138
26 : ie9k_iosxe.17.09.04.SPA.bin 596133776
27 : ix800-universalk9-bundle.SPA.156-3.M.bin 160968869
28 : ix800-universalk9-mz.SPA.155-3.M 61750062
29 : ix800-universalk9-mz.SPA.156-3.M 63753767
30 : ix800_yocto-1.7.2.tar 2877440
31 : ix800_yocto-1.7.2_python-2.7.3.tar 6912000
32 : ix8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
33 : pt1000-i-mz.122-28.bin 5571584
34 : pt3000-16q412-mz.121-22.EA4.bin 3117390
ftp>put applicationScenario2.txt

Writing file applicationScenario2.txt to ftp1.sea.com:
File transfer in progress...

[Transfer complete - 20 bytes]

20 bytes copied in 0.025 secs (800 bytes/sec)
ftp>
```

Figure B.2.3: Command Prompt FTP

Scenario 3 Screenshots

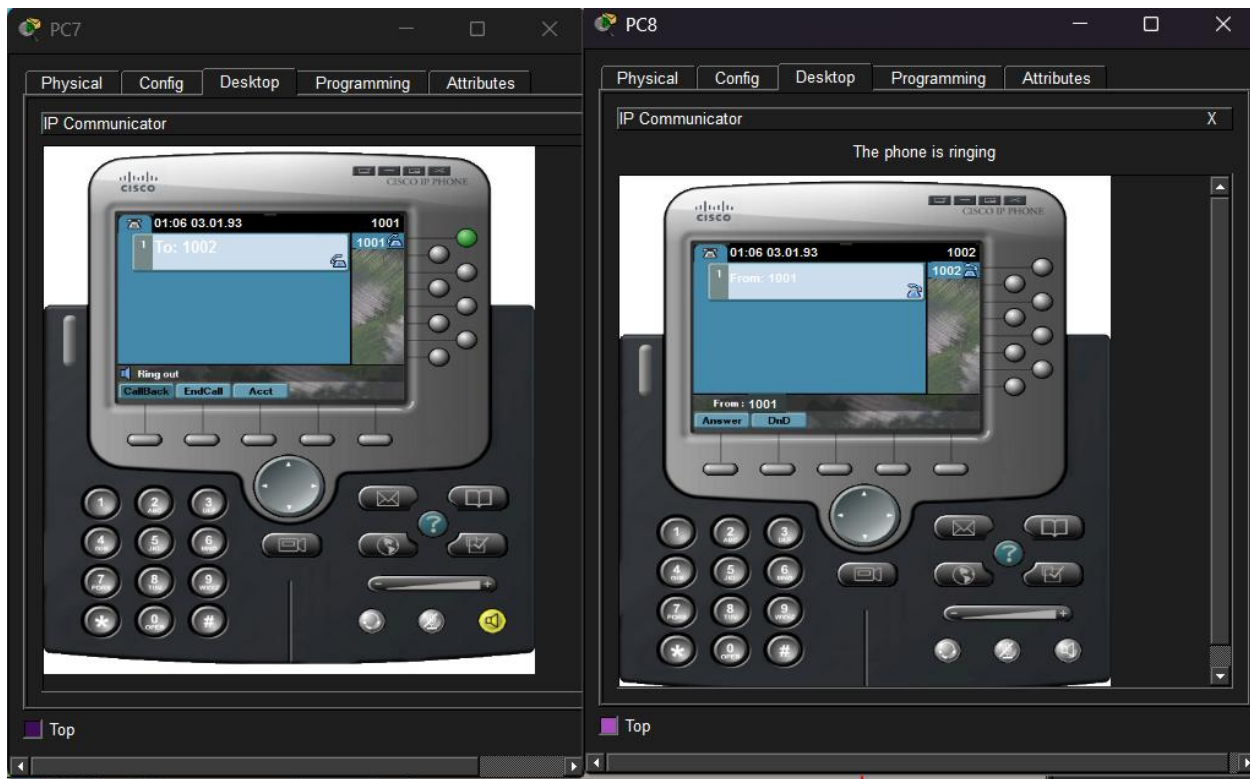


Figure B.3.1: IP Communicator

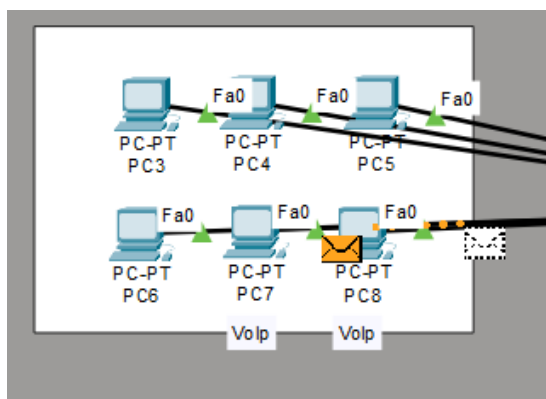


Figure B.3.2: Packet Transfer

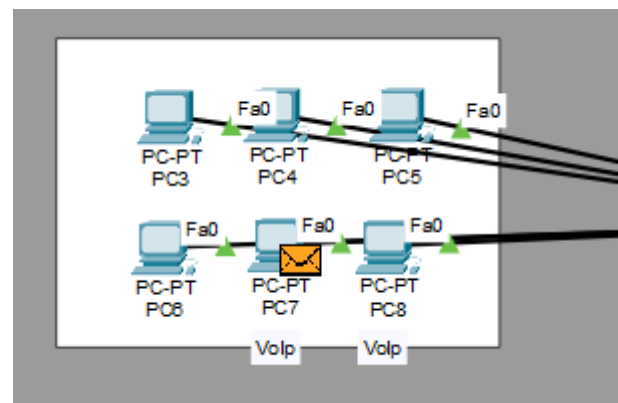


Figure B.3.3: Packet Transfer

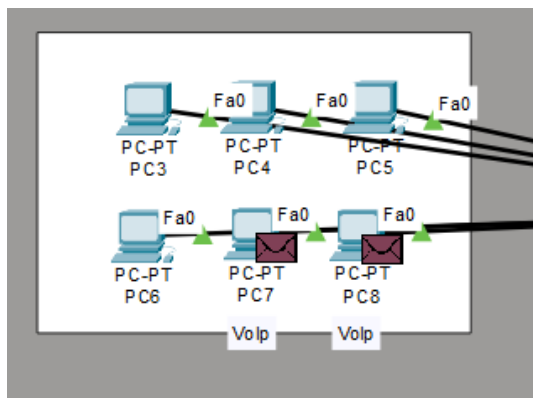


Figure B.3.4: Packet Transfer

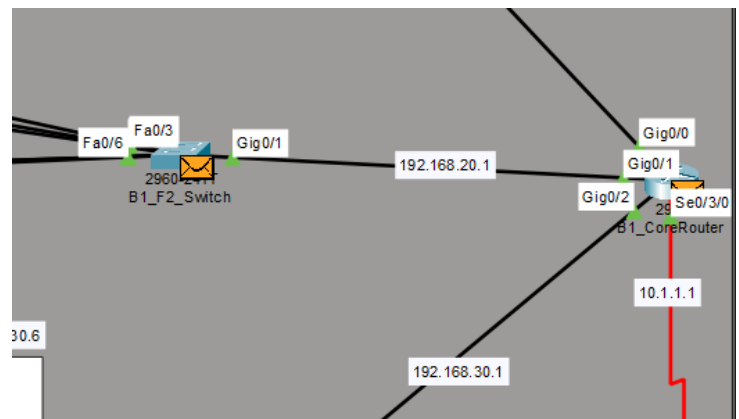


Figure B.3.5: Packet Transfer

Scenario 4 Screenshots

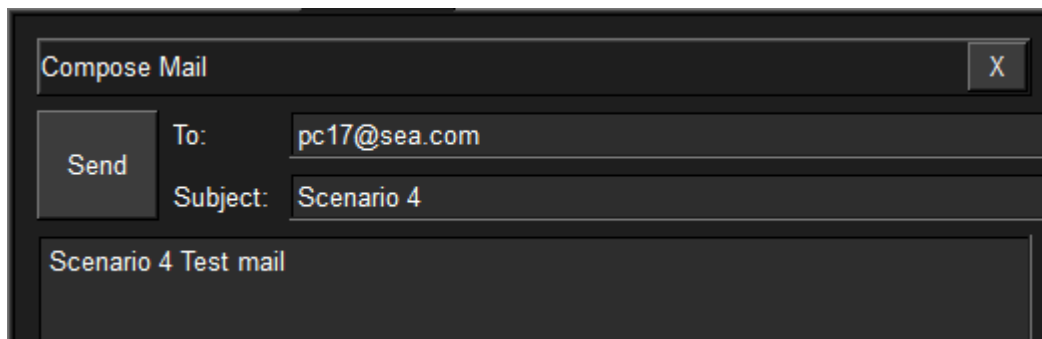


Figure B.4.1: Compose Mail

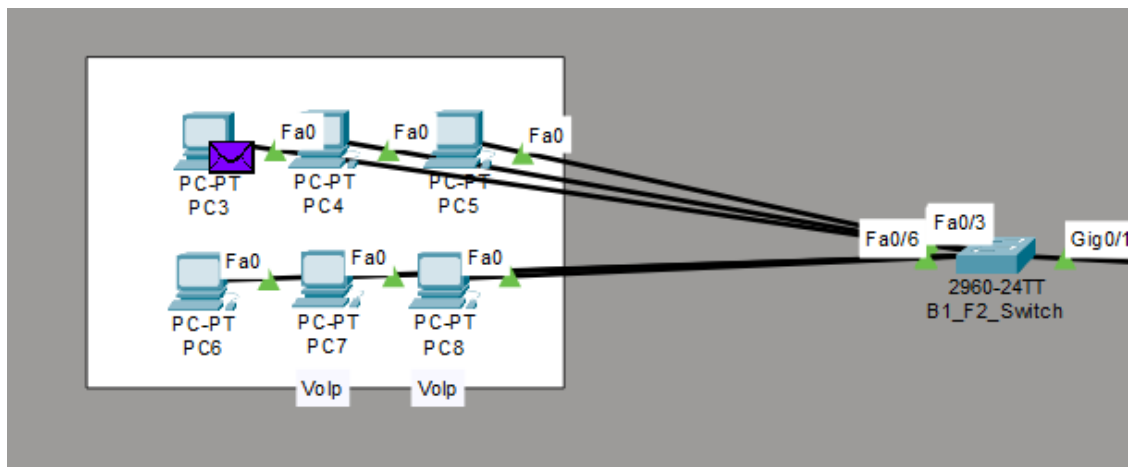


Figure B.4.2: Packet Transfer

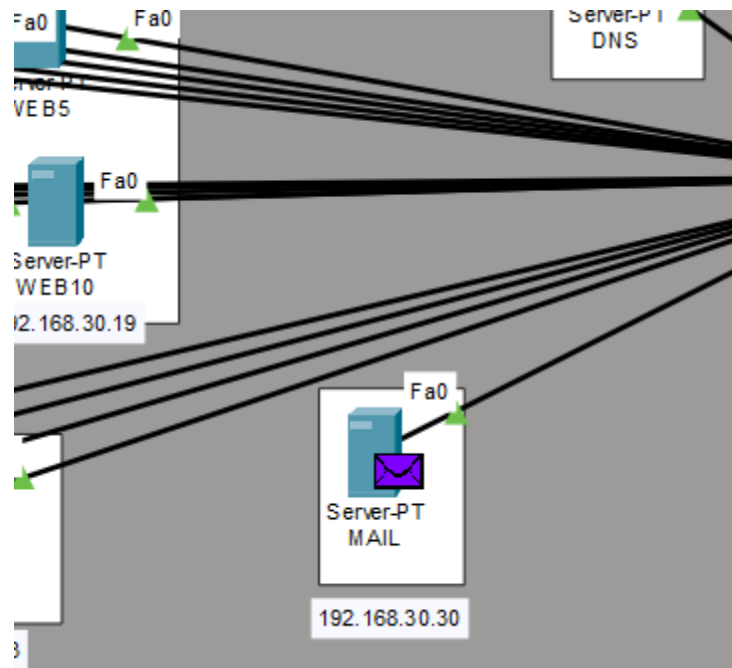


Figure B.4.3: Packet Transfer

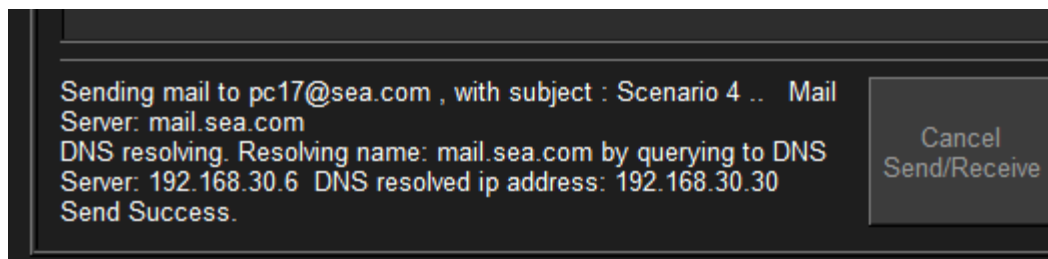


Figure B.4.4: Successful Mail

Scenario 5 Screenshots

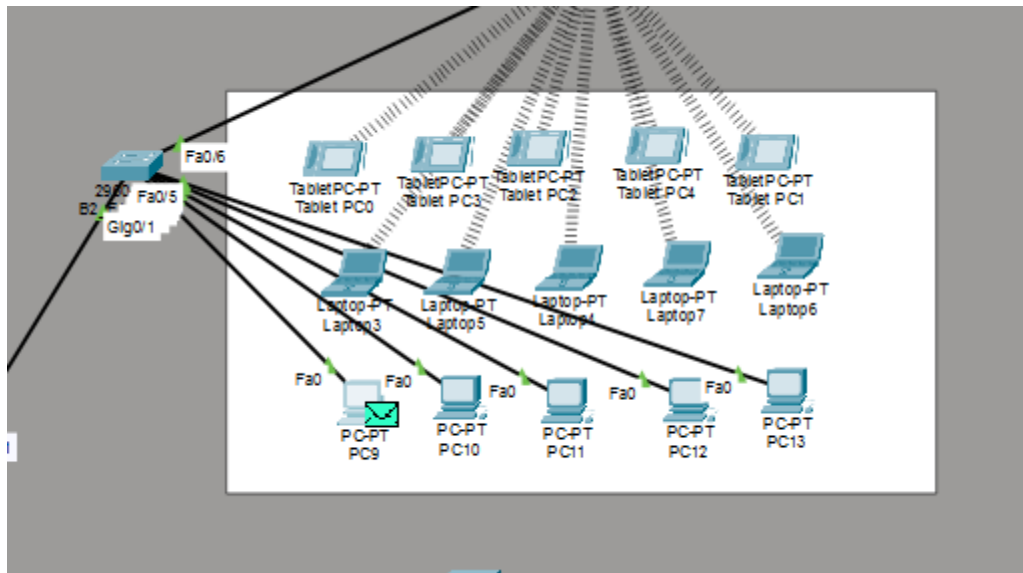


Figure B.5.1: Packet Transfer

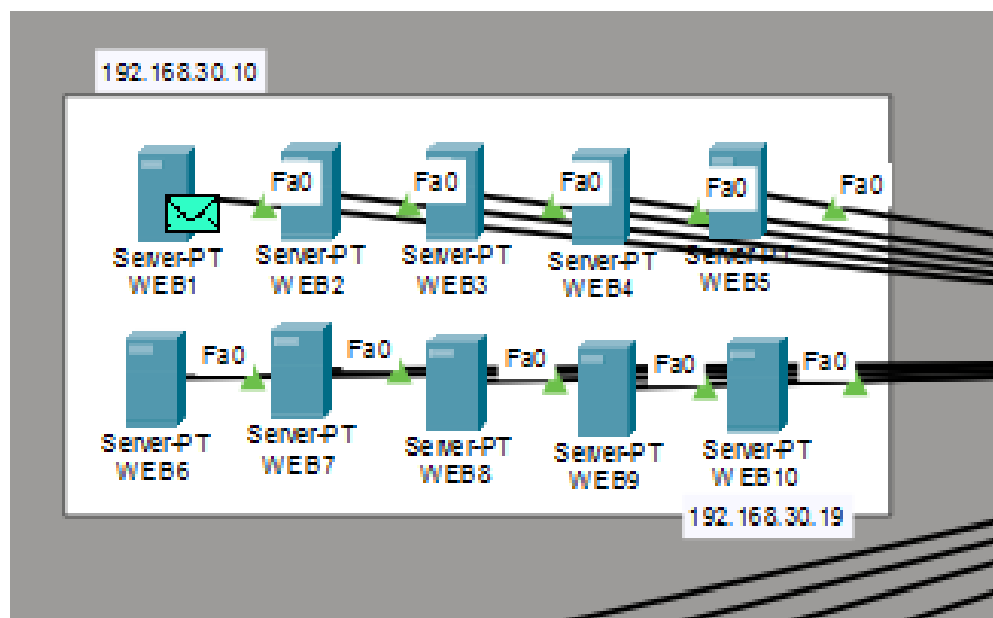


Figure B.5.2: Packet Transfer

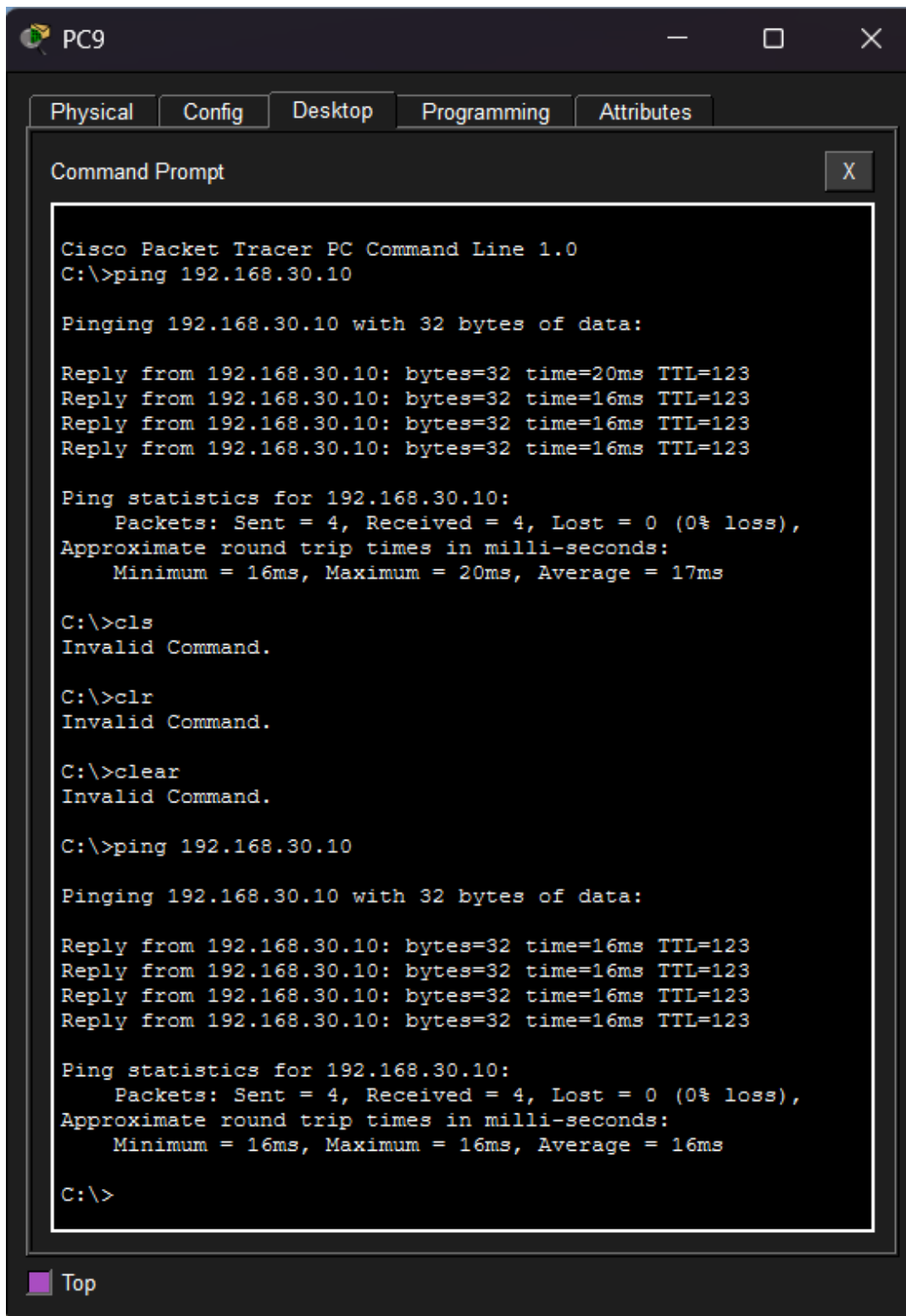


Figure B.5.1: Packet Transfer

Scenario 6 Screenshots

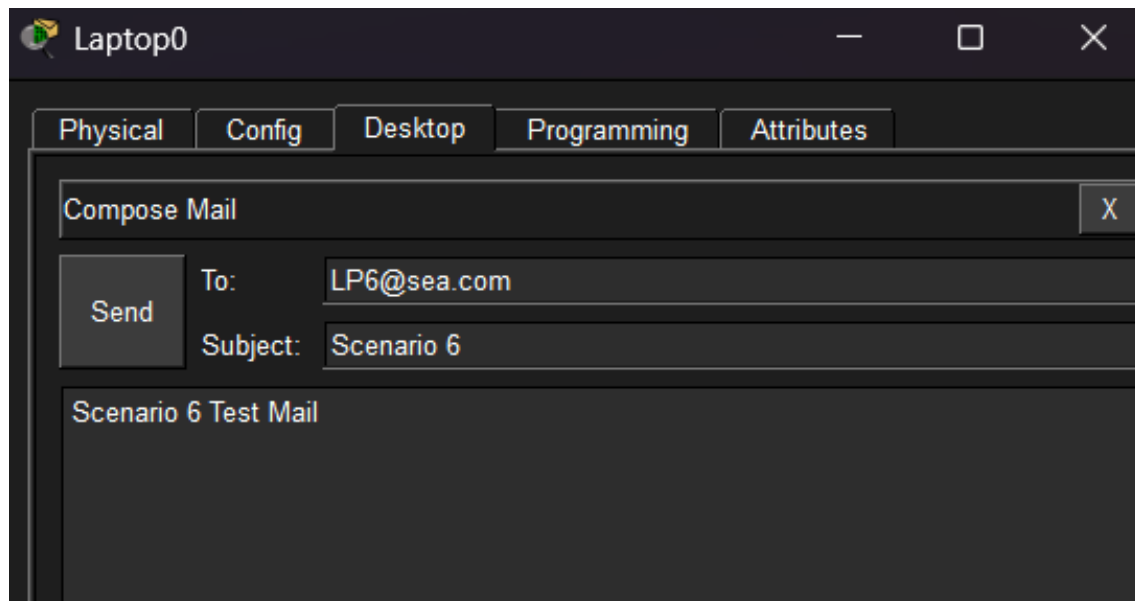


Figure B.6.1: Compose Mail

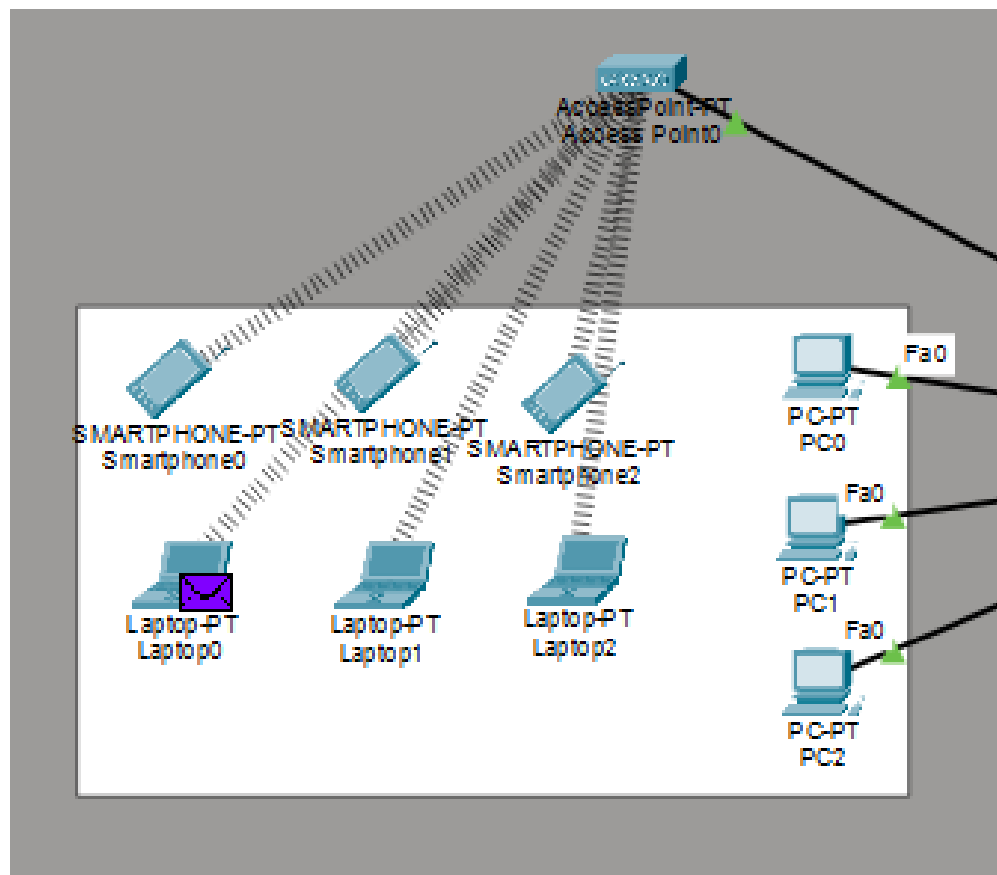


Figure B.6.2: Packet Transfer

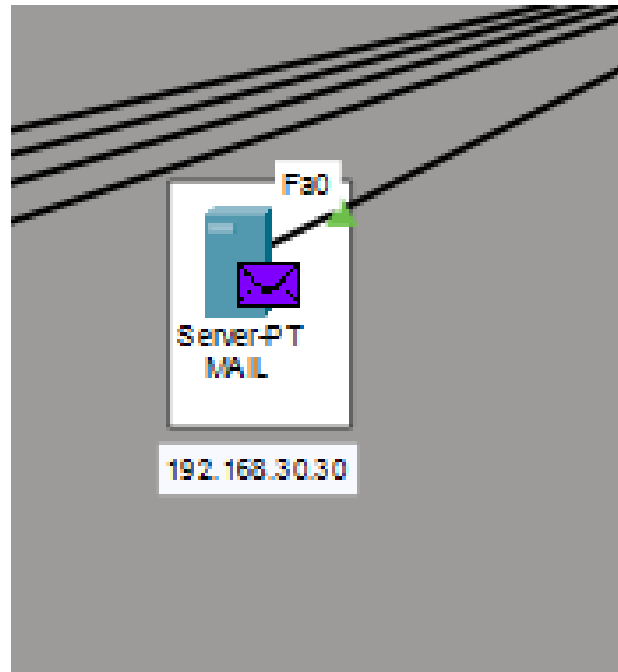


Figure B.6.3: Packet Transfer

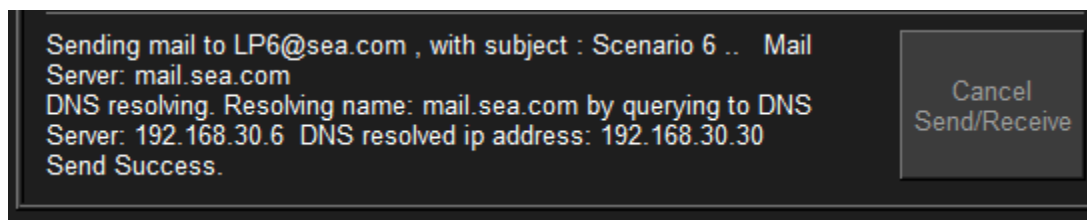


Figure B.6.4: Successful Mail

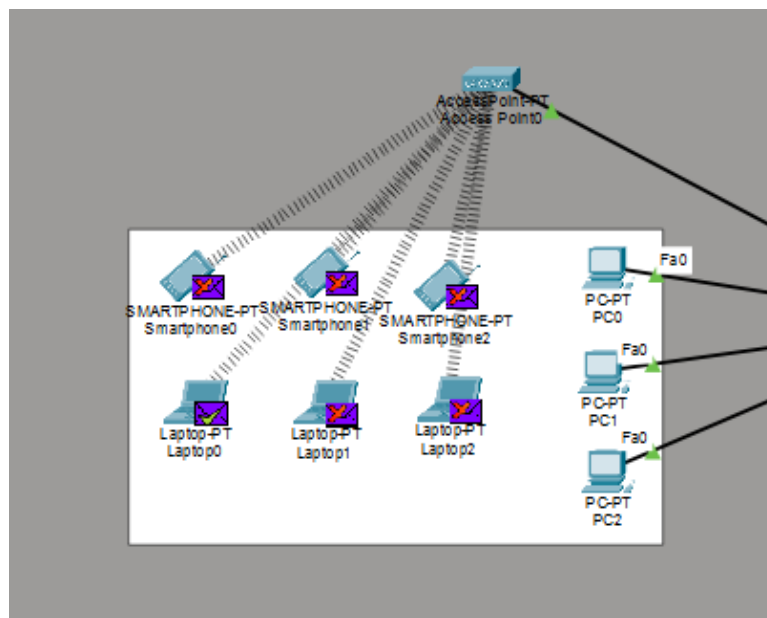


Figure B.6.5: Packet Transfer

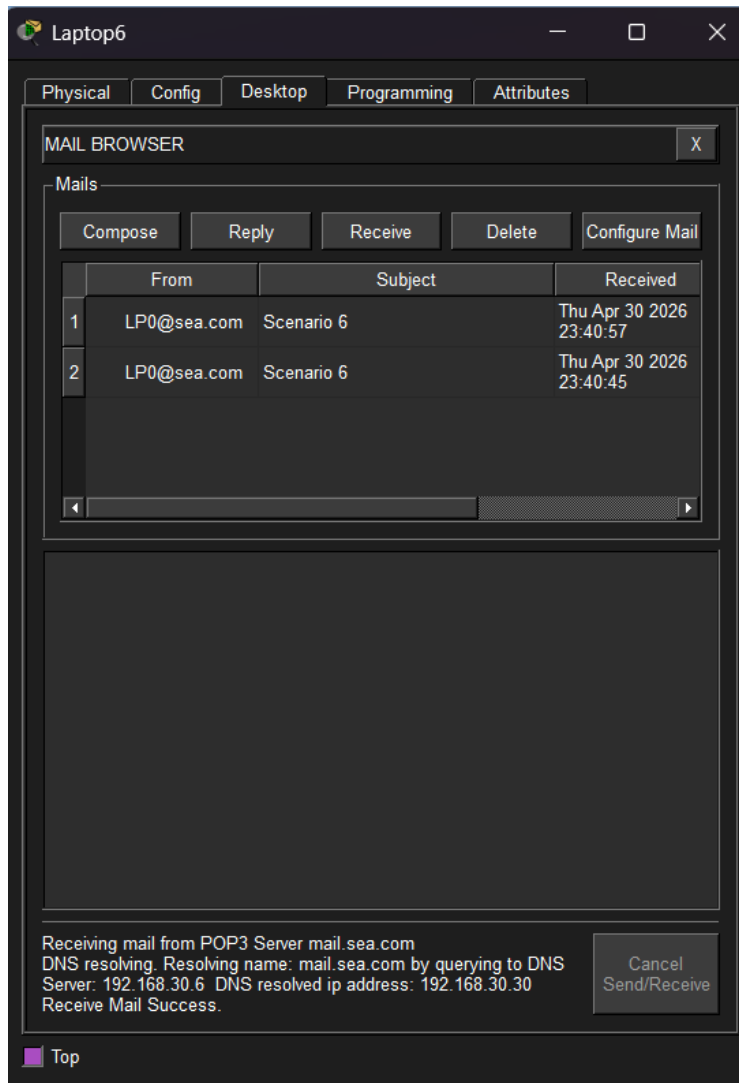


Figure B.6.6: Received Mails

Scenario 7 Screenshots

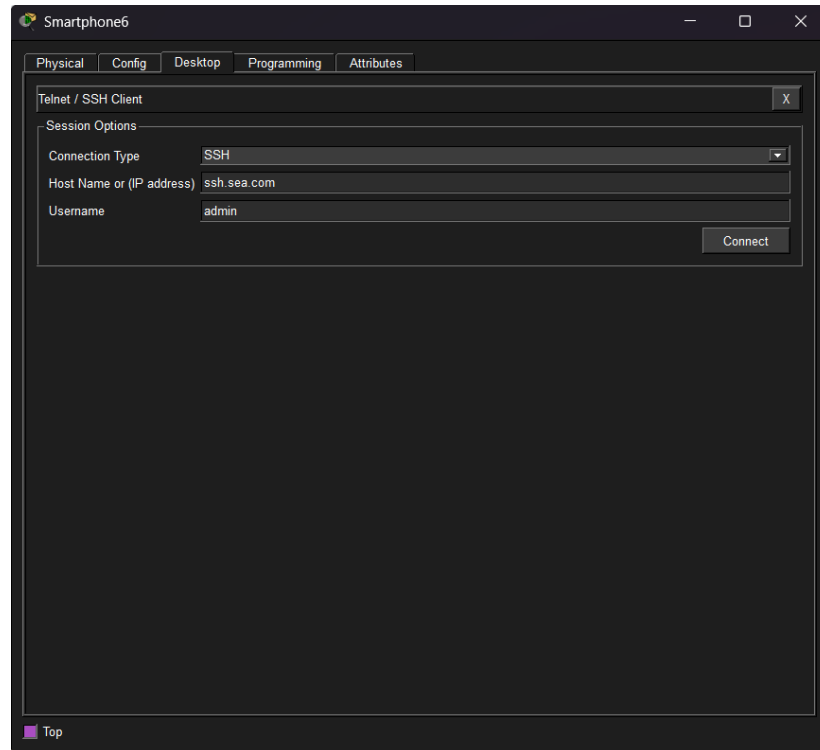


Figure B.7.1: SSH Connection

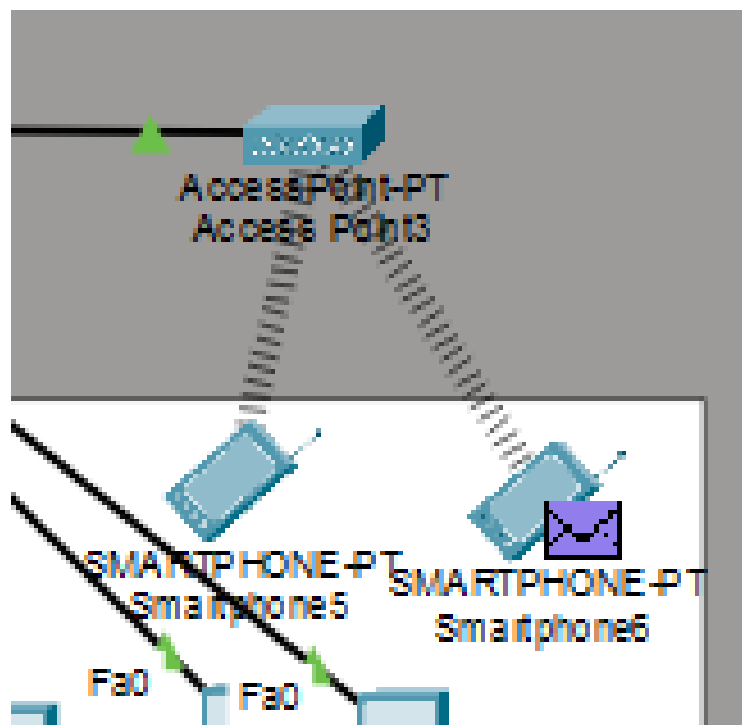


Figure B.7.2: Packet Transfer

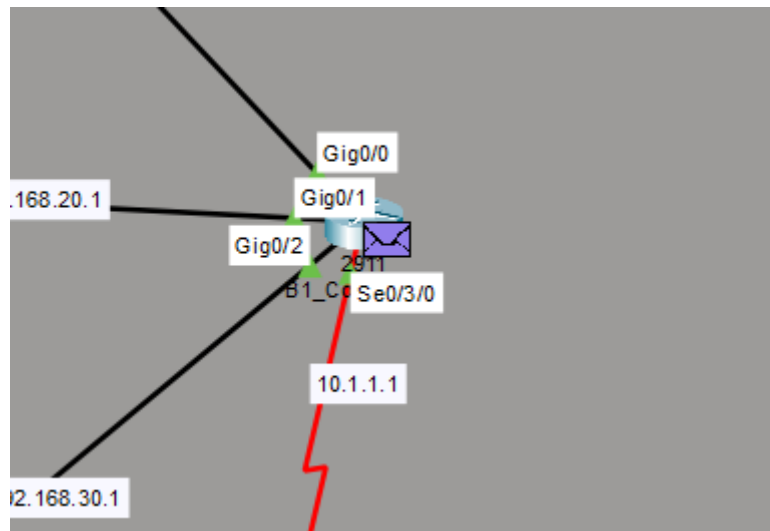


Figure B.7.3: Packet Transfer

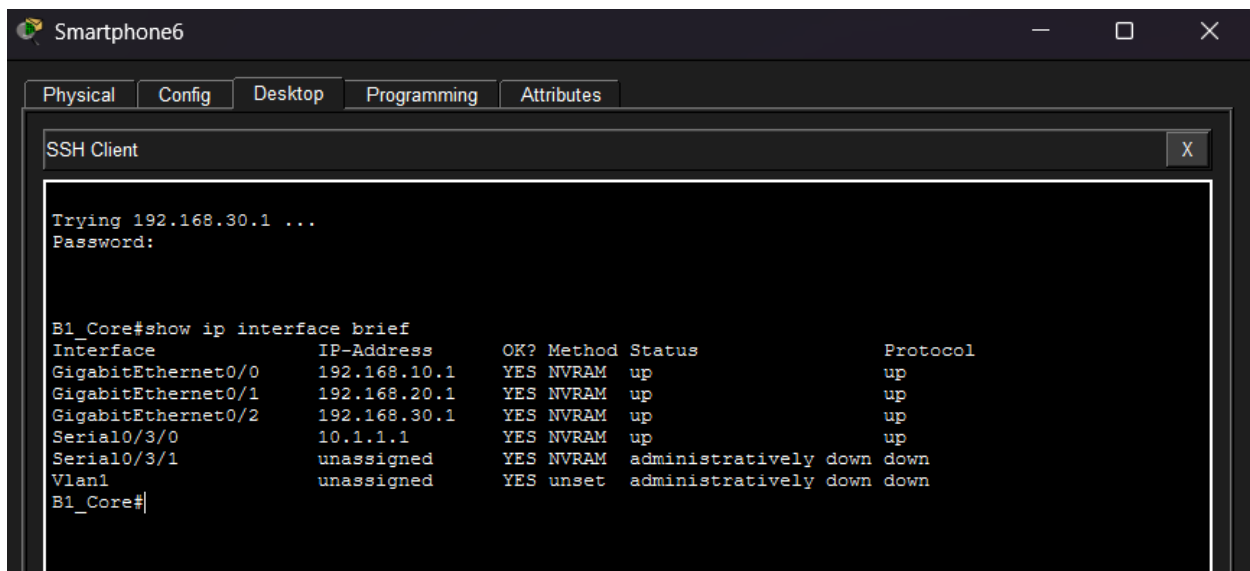
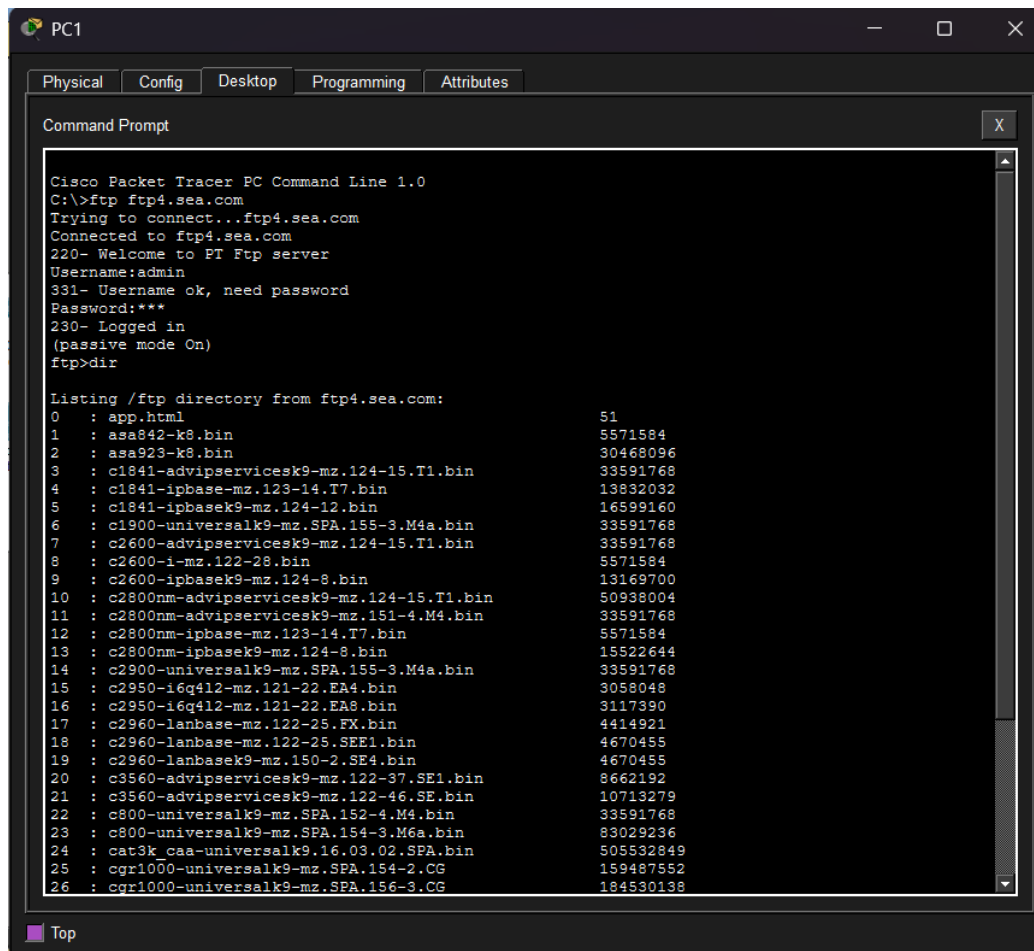


Figure B.7.4: SSH Connection to Router

Scenario 8 Screenshots (Custom)



The screenshot shows a PC1 window with a Command Prompt. The prompt displays the output of an FTP session. The user has connected to ftp4.sea.com as 'admin' and is now in the 'ftp>' directory listing mode. The listing shows a directory of files with their names and sizes.

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp ftp4.sea.com
Trying to connect...ftp4.sea.com
Connected to ftp4.sea.com
220- Welcome to PT Ftp server
Username:admin
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from ftp4.sea.com:
 0 : app.html                               51
 1 : asa842-k8.bin                         5571584
 2 : asa923-k8.bin                         30468096
 3 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
 4 : c1841-ibase-mz.123-14.T7.bin          13832032
 5 : c1841-ibasek9-mz.124-12.bin           16599160
 6 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
 7 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
 8 : c2600-i-mz.122-28.bin                 5571584
 9 : c2600-ibasek9-mz.124-8.bin            13169700
10 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
11 : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
12 : c2800nm-ibase-mz.123-14.T7.bin        5571584
13 : c2800nm-ibasek9-mz.124-8.bin          15522644
14 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
15 : c2950-16q412-mz.121-22.EA4.bin       3058048
16 : c2950-16q412-mz.121-22.EA8.bin       3117390
17 : c2960-lanbase-mz.122-25.FX.bin        4414921
18 : c2960-lanbase-mz.122-25.SEE1.bin      4670455
19 : c2960-lanbasek9-mz.150-2.SE4.bin       4670455
20 : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
21 : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
22 : c800-universalk9-mz.SPA.152-4.M4.bin   33591768
23 : c800-universalk9-mz.SPA.154-3.M6a.bin  83029236
24 : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
25 : cgr1000-universalk9-mz.SPA.154-2.CG   159487552
26 : cgr1000-universalk9-mz.SPA.156-3.CG   184530138
```

Figure B.8.1: FTP Connection

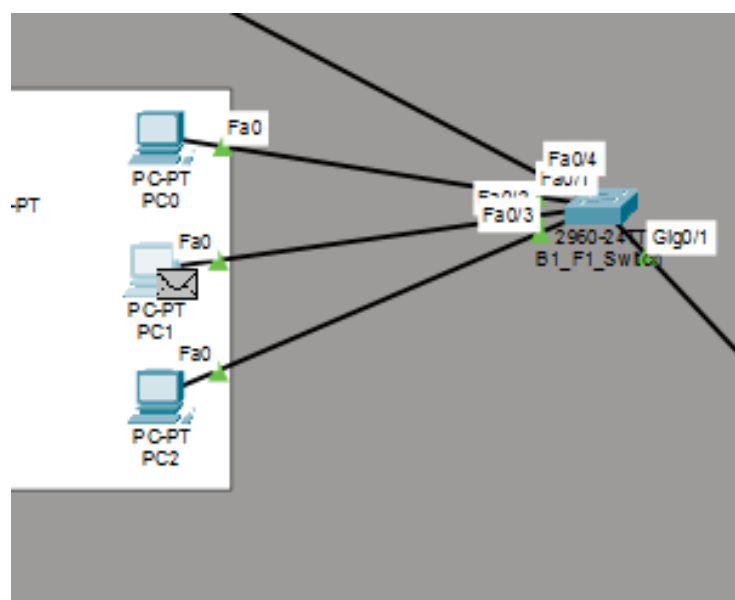


Figure B.8.2: Packet Transfer

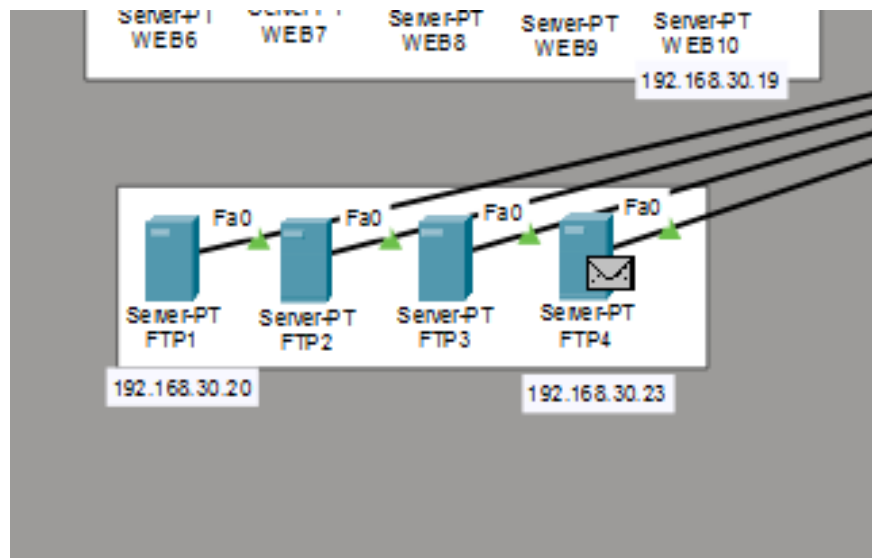


Figure B.8.3: Packet Transfer

```
ftp>get app.html

Reading file app.html from ftp4.sea.com:
File transfer in progress...

[Transfer complete - 51 bytes]

51 bytes copied in 0.016 secs (3187 bytes/sec)
ftp>
```

Figure B.8.4: Downloading a File

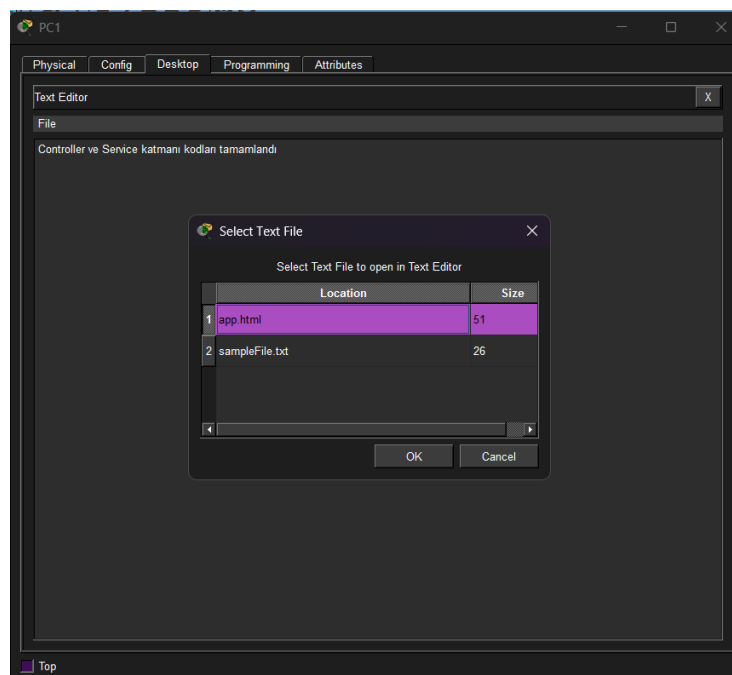


Figure B.8.5: Downloaded File

Scenario 9 Screenshots (Custom)

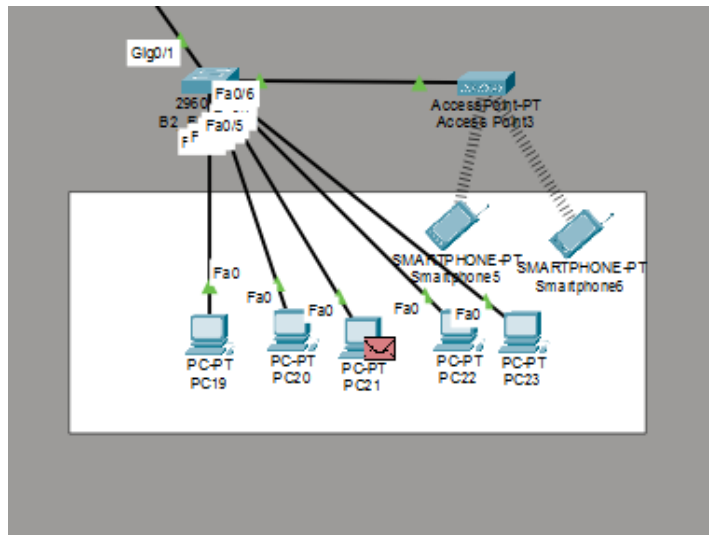


Figure B.9.1: Packet Transfer

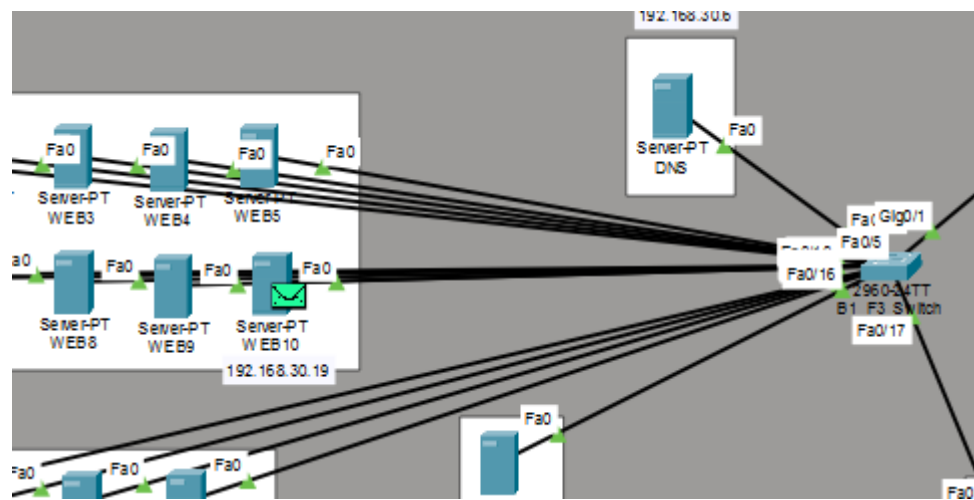
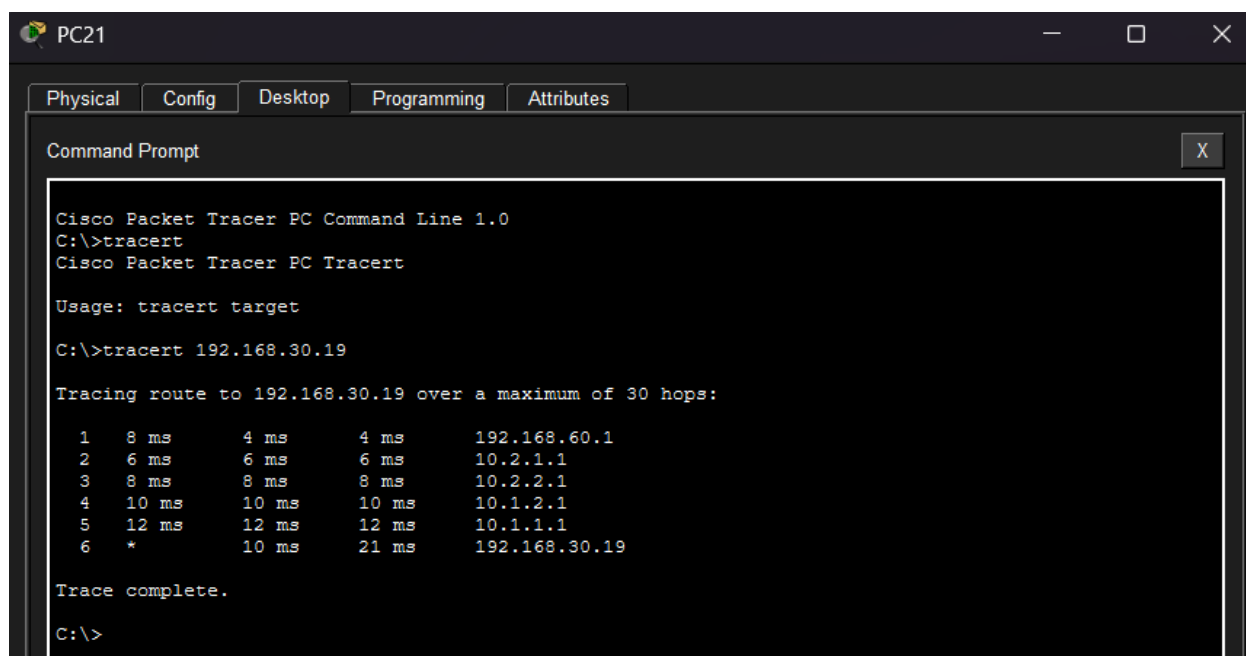


Figure B.9.2: Packet Transfer



The image shows a screenshot of a PC21 window. The window has a title bar with 'PC21' and standard minimize, maximize, and close buttons. Below the title bar are five tabs: 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Config' tab is selected. Inside the window is a 'Command Prompt' window with a black background and white text. The text in the Command Prompt shows the execution of a traceroute command in Cisco Packet Tracer.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert
Cisco Packet Tracer PC Tracert

Usage: tracert target

C:\>tracert 192.168.30.19

Tracing route to 192.168.30.19 over a maximum of 30 hops:

  1  8 ms      4 ms      4 ms      192.168.60.1
  2  6 ms      6 ms      6 ms      10.2.1.1
  3  8 ms      8 ms      8 ms      10.2.2.1
  4 10 ms     10 ms     10 ms     10.1.2.1
  5 12 ms     12 ms     12 ms     10.1.1.1
  6  *         10 ms     21 ms     192.168.30.19

Trace complete.

C:\>
```

Figure B.9.3: Completed Trace Route

Appendix C: Device Configuration Listings

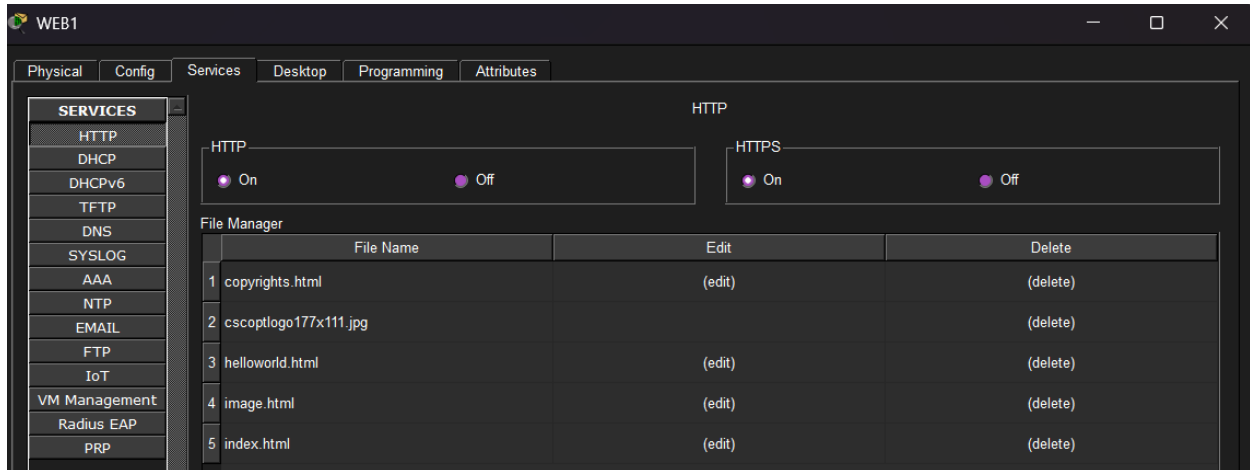


Figure C.1: Web Server Configuration

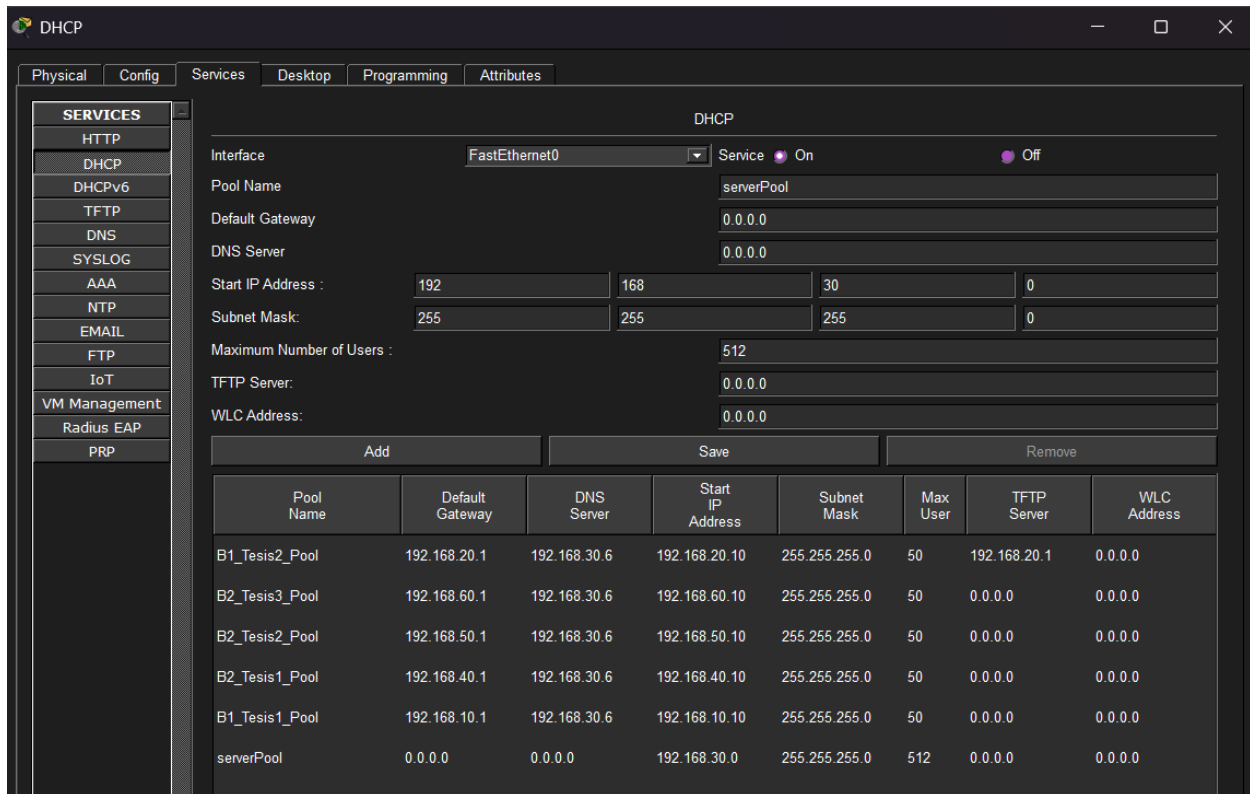


Figure C.2: DHCP Server Configuration

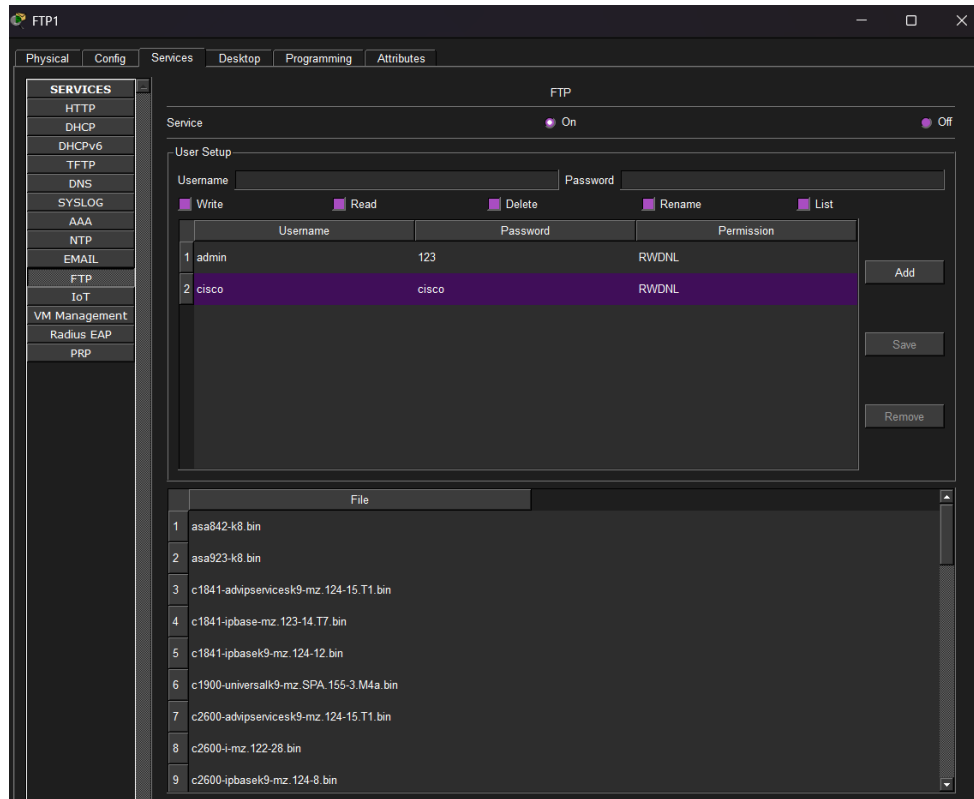


Figure C.3: FPT Server 1 Configuration

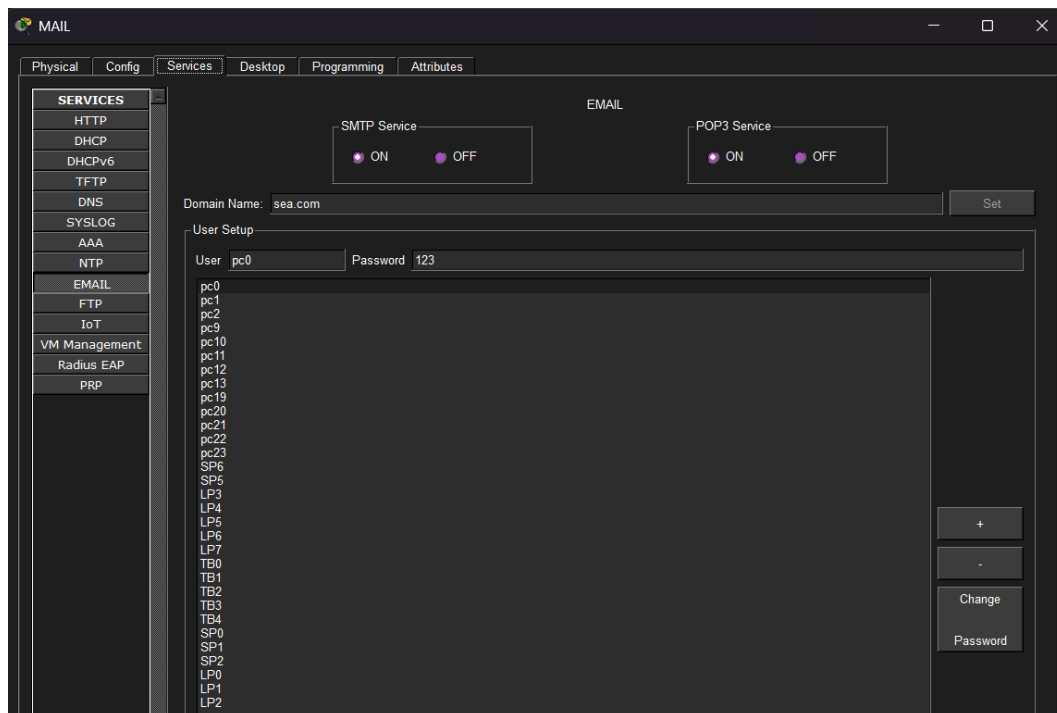


Figure C.4: Mail Server Configuration

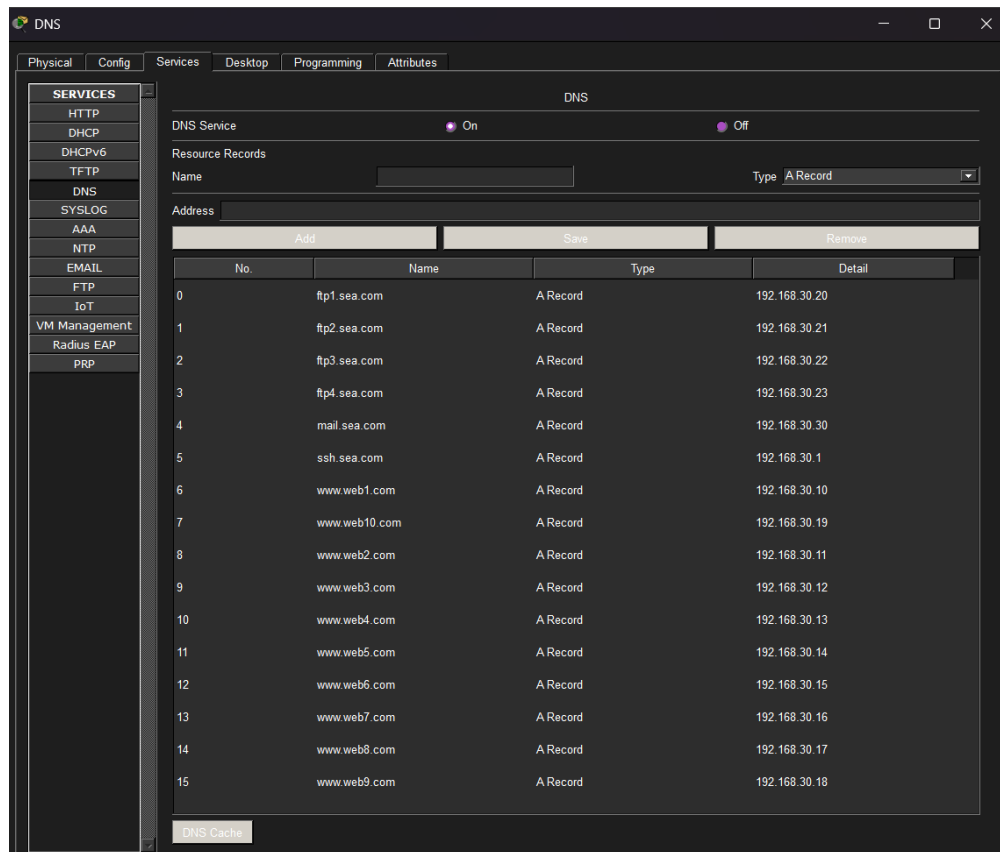


Figure C.5: DNS Server Configuration

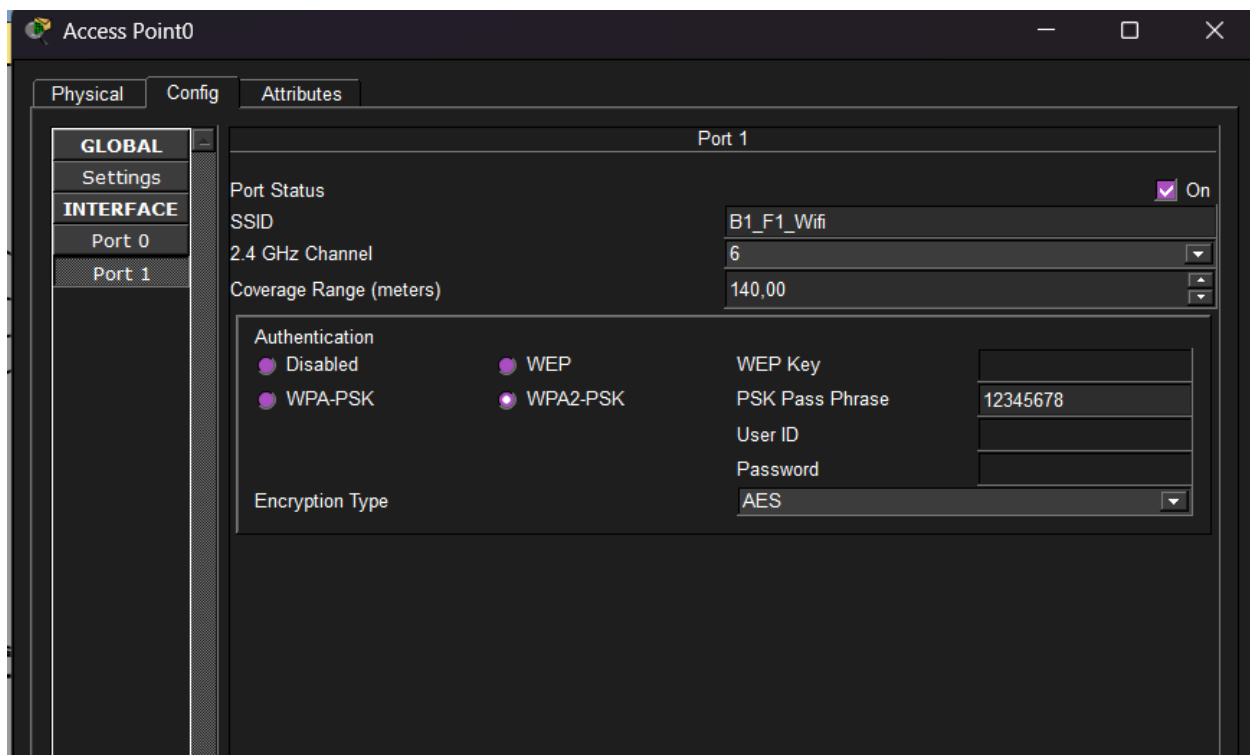


Figure C.6: Access Point Configuration